Semantic Directions Emerge from Sparse Token Mixtures

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Abstract

Hidden representations of large language models (LLMs) encode diverse semantic attributes—sentiment, truthfulness, safety, language identity—in linearly separable form, implying the existence of *semantic directions* that partition the representation space along each attribute. These directions underpin representation engineering and activation steering, yet remain opaque: as dense vectors spanning thousands of dimensions, *what* constitutes a semantic direction—which components give rise to it—is poorly understood.

We introduce **Token-Mixture**, a framework that constrains the weight vector of a linear probe to lie in the row space of the LM head's token embedding matrix as a non-negative sparse combination. Under this constraint, the decision boundary decomposes from an abstract high-dimensional vector into an interpretable mixture of specific token embeddings.

Across four LLMs and four semantic classification tasks, a mere 100 tokens—0.07% of the full vocabulary—recover 96% of the unconstrained linear probe performance. This sparse recovery is structurally specific to the token embedding basis: under the same budget, random token selection (64%) or substitution with an arbitrary orthonormal basis (77%) yields sharp performance collapse. Different semantic tasks activate nearly disjoint token sets (cross-task Jaccard ≈ 0), revealing that the vocabulary space is partitioned with respect to each semantic concept. Knockout ablation and activation steering experiments further confirm that the selected anchor tokens are causal constituents of the semantic direction.

Token-Mixture decomposes what were previously opaque thousand-dimensional vectors into readable token inventories—e.g., a "truthfulness direction" spanned by tokens such as `honest`, `verify`, and `fact`. Token embeddings serve as a privileged sparse basis for semantic directions, and this structure reproduces consistently across models and tasks.

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1 INTRODUCTION

The hidden representations of large language models (LLMs) encode diverse semantic attributes—sentiment, truthfulness, safety, and language identity—in linearly separable form [Alain and Bengio, 2017, Belinkov, 2022]. This finding implies the existence of *semantic directions* in hidden space that distinguish each attribute, forming the theoretical basis for representation engineering and activation steering [Zou et al., 2023, Turner et al., 2023].

Yet these semantic directions remain opaque arrows in a high-dimensional space. We know that such directions *exist*, but what they *are*—what components constitute them and why they point in a particular direction—remains unknown. The weight vector $\mathbf{w} \in \mathbb{R}^d$ learned by a standard linear probe is a dense vector spanning thousands of dimensions, and no method has been proposed to decompose it into human-interpretable units.

This work poses the following question:

> **Can semantic directions be read in terms of the model's own vocabulary?**

Our approach is straightforward. Instead of learning the probe weight as an unconstrained vector in $\mathbb{R}^d$, we restrict it to the row space of the LM head's token embedding matrix $\mathbf{T} \in \mathbb{R}^{V \times d}$:

$$\mathbf{w} = \mathbf{T}^\top \boldsymbol{\alpha}, \quad \boldsymbol{\alpha} \geq 0 \tag{1}$$

Under this constraint, the semantic decision boundary is no longer an abstract high-dimensional vector but a non-negative linear combination of specific token embeddings—that is, an interpretable list of tokens. By further imposing

a global top-K constraint that activates only K entries of $\boldsymbol{\alpha}$, we can directly measure the size of the minimal vocabulary set that supports a semantic direction. We call this framework **Token-Mixture**.

For this reformulation to be meaningful, two conditions must hold. First, the vocabulary constraint must not excessively sacrifice performance—if the cost of being "confined" to the vocabulary space is large, the approach is merely a poor approximation. Second, the observed sparsity must be specific to the token embedding coordinate system—if the same level of sparse recovery is achievable in an arbitrary coordinate system, it would reflect the general low-dimensionality of the embedding space rather than the semantic structure of the vocabulary.

Through systematic experiments across four LLMs and four semantic classification tasks, we show that both conditions are met. Moreover, we find that this vocabulary-level sparsity reveals interpretable and systematic structure.

(1) Extreme vocabulary-level sparsity. With only $K{=}100$ tokens—0.07% of the full vocabulary—Token-Mixture recovers 96% of the performance of an unconstrained linear probe on average. This indicates that semantic directions admit extremely compressed representations over the vocabulary.

(2) Specificity of the token embedding coordinate system. Under the same $K{=}100$ budget, randomly selecting tokens degrades performance to 64%, and replacing the coordinate system with a random orthonormal basis yields a similar collapse. Sparse recovery is not a general phenomenon that arises in arbitrary coordinate systems; it is a structural phenomenon specific to the token embedding basis.

(3) Concept-specific lexical support. Different semantic tasks activate nearly disjoint token sets (cross-task Jaccard ≈ 0). The anchors for sentiment and language occupy entirely different regions of the vocabulary. This shows that the vocabulary space is not a flat space sharing universally important tokens, but rather a partitioned structure in which each semantic concept occupies a distinct region.

(4) Vocabulary-level quantification of semantic complexity. The number of tokens required to reach 95% of full-vocabulary performance varies systematically across tasks: language classification saturates at $K{=}100$, whereas truthfulness classification requires $K{=}11{,}000$. Tasks that rely on surface-level linguistic cues saturate rapidly with very few tokens, while tasks requiring compositional semantic reasoning demand broader lexical support. This **saturation hierarchy** provides a novel measure for quantifying semantic complexity in the vocabulary dimension.

Taken together, this work shifts the unit of interpretation for semantic directions. What were previously opaque vectors spanning thousands of dimensions can now be decomposed, via Token-Mixture, into readable token lists such as "the `honest`, `verify`, `fact` direction." This decomposition is possible because token embeddings serve as a **privileged sparse basis** for semantic directions, and this structure is consistently reproduced across models and tasks.

Our contributions are as follows:

- We propose the Token-Mixture framework, which constrains semantic directions to the vocabulary row space.
- We demonstrate, across diverse models and tasks, that semantic decision boundaries exhibit extreme vocabulary-level sparsity.
- We analyze the per-task lexical support structure and saturation hierarchy, showing that the vocabulary space functions as an interpretable coordinate system for semantic concepts.
- Through knock-out ablation and activation steering, we causally verify that the identified anchor tokens are functionally essential components of semantic directions.

2 RELATED WORK

Linear Probes and Semantic Directions. Linear probing has established that diverse semantic attributes—such as syntax, sentiment, and truthfulness—are linearly separable in LLM hidden representations [Alain and Bengio, 2017, Hewitt and Manning, 2019, Radford et al., 2017, Azaria and Mitchell, 2023, Burns et al., 2023, Belinkov and Glass, 2019, Tenney et al., 2019]. However, conventional probes learn unconstrained weight vectors, revealing that a semantic direction exists without clarifying which vocabulary elements support it. We instead restrict the probe to the vocabulary row space, enabling direct inspection of its token-level support.

Vocabulary Projection and Logit-Based Interpretability. Logit Lens and Tuned Lens project hidden states through the LM head to examine token prediction distributions [nostalgebraist, 2020, Belrose et al., 2023]. While these methods analyze representations via vocabulary space, they focus on token predictions of hidden states. In contrast, we decompose the semantic decision boundary itself into a sparse combination of vocabulary tokens.

Sparse Representation and Dictionary Learning. Sparse Autoencoders and related approaches extract interpretable features from hidden activations using learned or external dictionaries [Elhage et al., 2022, Cunningham et al., 2024, Marks et al., 2025, Bhalla et al., 2024]. Our method similarly enforces sparsity, but uses the model's own output vocabulary as the dictionary, providing a supervised decom-

position of semantic directions into readable token-level units.

3 METHOD

This section presents the Token-Mixture framework for reading semantic directions as sparse combinations of tokens over the model's own vocabulary. We aim to answer the following questions:

(Q1) Can a semantic decision boundary be reconstructed from only a small number of tokens in the vocabulary row space?

(Q2) If so, is this sparse recovery specific to the token embedding coordinate system, or is it a general phenomenon that holds in arbitrary coordinate systems?

(Q3) Do different semantic concepts activate different token sets?

To this end, we constrain the weight of a standard linear probe to the model's output vocabulary row space and define a *vocabulary-constrained semantic reconstruction* problem in which the number of active tokens is systematically controlled.

3.1 VOCABULARY-CONSTRAINED SEMANTIC RECONSTRUCTION

Unconstrained Linear Probe. Let $\mathbf{h}(x) \in \mathbb{R}^d$ denote the hidden representation extracted from a specific layer of a frozen LLM. A standard linear probe directly learns a class-specific weight matrix $\mathbf{W} \in \mathbb{R}^{d \times C}$:

$$\text{logits}(x) = \mathbf{h}(x)^\top \mathbf{W} + \mathbf{b}. \tag{2}$$

This setup can verify the *existence* of semantic separability, but the learned $\mathbf{W}$ remains a dense vector in d dimensions. While we can confirm that a direction exists, *what* that direction is—what components constitute it—remains opaque.

Row-Space Restriction. Let $\mathbf{T} \in \mathbb{R}^{V \times d}$ denote the token embedding matrix of the LM head. We restrict the weight matrix to the vocabulary row space:

$$\mathbf{W} = \mathbf{T}^\top \boldsymbol{\alpha}, \qquad \boldsymbol{\alpha} \in \mathbb{R}^{V \times C}. \tag{3}$$

Under this constraint, a semantic direction is no longer an abstract high-dimensional vector but a linear combination of token embeddings—that is, an interpretable list of tokens. Each entry of $\boldsymbol{\alpha}$ directly indicates how much the corresponding token contributes to the semantic axis, making it possible to read off the tokens that constitute the direction.

Substituting into the logits expression yields

$$\text{logits}(x) = (\mathbf{T}\mathbf{h}(x))^\top \boldsymbol{\alpha} + \mathbf{b}. \tag{4}$$

Here, $\mathbf{T}\mathbf{h}(x) \in \mathbb{R}^V$ is the projection of the hidden state into vocabulary space. This is the same vocabulary projection operation used by Logit Lens, but our objective differs: rather than interpreting the token prediction distribution of individual hidden states, we analyze the learned $\boldsymbol{\alpha}$ to determine *which tokens compose* the semantic direction.

Constructive Support Constraint. To clearly define the set of tokens that constitute a semantic direction, we introduce the constraint $\boldsymbol{\alpha} \geq 0$.

If negative weights are permitted, opposing tokens can cancel each other to form the same boundary, making it difficult to answer the question "which tokens support this direction?" unambiguously. The non-negativity constraint forces each token to make only a positive contribution to the semantic direction, allowing the set of active tokens to be interpreted as the *constructive support set* of that direction.

3.2 MINIMAL LEXICAL SUPPORT ESTIMATION

The central question is: "how many tokens suffice to read off a semantic direction?" To measure this quantitatively, we introduce a global top-K constraint that activates only K tokens out of the full vocabulary V.

We define the aggregate contribution of each token i as

$$s_i = \sum_c \alpha_{i,c} \tag{5}$$

and retain only the active set $\mathcal{A}_K$ consisting of the top-K tokens. The coefficients for all remaining tokens are fixed to zero.

By systematically varying K, we can directly measure the *minimum number of tokens* required to recover a semantic direction. The resulting saturation curve serves as a measure of how concentrated each semantic concept is over the vocabulary.

4 EXPERIMENTAL SETUP

This section describes the experimental design for verifying whether the proposed Token-Mixture framework produces consistent results across diverse models and tasks. The experiments are organized around controlled comparisons corresponding to the questions (Q1–Q3) posed in §3.

4.1 TASKS

We evaluate four classification tasks spanning different semantic granularities.

Language (5-class). English, Spanish, Chinese, French, and German classification using XNLI [Conneau et al., 2018] (train 25K, val 25K).

Alignment (2-class). Jailbreak vs. benign query classification using JailBreakV-28k [Luo et al., 2024] (train 25K, val 5K).

Sentiment (2-class). Positive vs. negative classification on IMDB [Maas et al., 2011] (train 25K, val 25K).

Truthfulness (2-class). Truthful vs. false classification on TruthfulQA [Lin et al., 2022] (train 1,361, val 273).

4.2 MODELS

We use four instruction-tuned LLMs that differ in architecture, hidden dimension, and vocabulary size:

- Llama-3.1-8B-Instruct (d=4096, 32 layers, V=128K)
- Qwen2.5-7B-Instruct (d=3584, 28 layers, V=152K)
- GPT-OSS-20B (d=6144, 24 layers, V=201K)
- Phi-4 (d=5120, 40 layers, V=100K)

By examining whether consistent results emerge across models that differ in scale, vocabulary size, and training data, we assess the generality of the findings.

For each input, we extract the hidden state $\mathbf{h}_l(x) \in \mathbb{R}^d$ at the last token position of the autoregressive model. Layer selection is performed based on validation accuracy, and all baselines follow the same procedure.

4.3 BASELINES

To systematically verify that the sparse structure discovered by Token-Mixture is specific to the token embedding coordinate system, unique to each semantic concept, and constitutive of the direction itself, we design controls corresponding to each question.

Q1: Is sparse recovery specific to the token coordinate system? (§4.4)

(B1) Random Token Mixture (Identity Contrast). We retain the token embedding coordinate system $\mathbf{T}$ but select K tokens uniformly at random instead of via top-K, then train under the same procedure. This distinguishes whether performance stems from the selection of specific tokens or from a mere dimensionality reduction effect.

(B2) Random Basis Sparse Probe (Basis Contrast). We replace the token embedding $\mathbf{T}$ with a random orthonormal basis $\mathbf{B} \in \mathbb{R}^{d\times d}$ and train under the constraint $\mathbf{W} = \mathbf{B}^\top \boldsymbol{\beta}$, $\|\boldsymbol{\beta}\|_0 \le K$. This directly tests whether sparse recovery is a phenomenon specific to the token coordinate system or a general property that holds in arbitrary coordinate systems.

(B3) Linear Probe (Unconstrained Upper Bound). A standard linear classifier with no constraints on $\mathbf{W} \in \mathbb{R}^{d\times C}$. This serves as a performance upper bound for assessing whether the sparsity constraint of Token-Mixture incurs excessive performance loss.

Q2: Do different concepts activate different tokens? (§4.5)

(B4) Cross-task Jaccard Similarity. We measure the Jaccard similarity between the anchor sets (K=100) of different tasks within the same model. This determines whether different semantic concepts share "universally important tokens" or each concept occupies a distinct token set.

(B5) Saturation K Analysis. We systematically vary the token budget K from 10 to 30,000 and measure, for each task, the minimum K (saturation K_{sat}) at which 95% of the full-vocabulary performance is reached. This tests whether the cross-task differences in K_{sat} reflect the scale of token support that each semantic concept requires.

Q3: Are the selected tokens constitutive of the direction? (§4.6)

(B6) Knock-out Ablation (Causal Necessity). We remove class-specific top tokens from the trained Token-Mixture (without re-training) and measure the resulting performance change, comparing against removal of the same number of random tokens. This directly tests whether the selected tokens are merely correlated with the semantic direction or are constitutive elements of the direction itself.

(B7) Anchor Intervention (Causal Sufficiency). Whereas (B6) tests whether the direction collapses upon token removal, this experiment addresses the converse question: *can a sparse combination of anchor tokens alone steer the model's output?*

We inject into the model's final hidden state $\mathbf{h}$ both the full dense direction $\mathbf{w}_{\text{full}}$ and a sparse anchor direction $\mathbf{w}_{\text{anchor}}$ composed of only a small number of token embeddings. The intervention strength λ is calibrated to achieve a target margin shift, producing the following modified hidden states:

$$\mathbf{h}_{\text{full}} = \mathbf{h} + \lambda \mathbf{w}_{\text{full}}, \quad \mathbf{h}_{\text{anchor}} = \mathbf{h} + \lambda \mathbf{w}_{\text{anchor}}$$

We measure the KL divergence and rank preservation between the output distributions induced by the two interventions. If $\mathbf{h}_{\text{anchor}}$ reproduces the output of $\mathbf{h}_{\text{full}}$, it implies

Table 1: **Semantic directions are recovered by a very small number of anchors in the token embedding coordinate system.** Top: Token-Mixture (Anchor) performance as a function of token budget K. At $K{=}100$, 96% of full-vocabulary performance is already reached. Bottom: Under the same $K{=}100$ constraint, randomly selecting tokens (Rand-Tok) or using a random orthonormal basis (Rand-Basis) causes a sharp performance drop. For each (model, task, K) combination, the optimal layer is independently selected based on validation accuracy.

		Token-Mixture					**Controls** ($K{=}100$)		
Task	**Model**	$K{=}10$	$K{=}100$	$K{=}1$K	$K{=}5$K	Full	Rand-Tok	Rand-Basis	LinProbe
Language	Llama	69.48	**97.00**	99.26	99.72	99.80	50.85	76.18	99.86
	Qwen	70.44	**98.56**	99.04	99.40	99.80	46.89	82.53	99.94
	GPT-OSS	53.74	**85.60**	94.18	95.04	93.34	43.77	74.08	94.00
	Phi-4	63.20	**97.56**	98.84	99.40	99.46	46.83	64.49	99.14
Alignment	Llama	85.42	**97.68**	99.84	99.96	99.96	91.87	95.85	99.98
	Qwen	80.96	**97.30**	99.52	99.70	99.74	78.90	93.17	99.70
	GPT-OSS	94.46	**99.08**	99.64	99.54	99.64	89.32	96.39	99.86
	Phi-4	93.26	**99.12**	99.76	99.92	99.94	94.13	96.09	99.94
Sentiment	Llama	76.65	**89.04**	92.29	92.94	93.71	71.76	79.36	92.57
	Qwen	71.88	**82.96**	88.27	89.82	89.60	60.22	68.67	91.52
	GPT-OSS	75.72	**80.77**	85.99	83.40	80.36	61.16	70.84	86.32
	Phi-4	71.59	**83.34**	90.90	92.78	90.65	67.40	73.46	93.50
TruthfulQA	Llama	63.74	**75.46**	78.75	81.68	86.45	61.54	71.31	80.59
	Qwen	60.44	**64.10**	69.96	75.82	72.89	52.87	64.35	76.56
	GPT-OSS	56.78	**62.27**	63.37	64.84	63.00	51.94	60.32	58.97
	Phi-4	63.74	**68.86**	75.46	79.85	78.02	57.60	64.71	78.02
Average		71.9	**86.2**	89.2	91.3	91.1	64.2	77.0	90.7

that the high-dimensional dense direction is functionally equivalent to a sparse combination of tokens.

(B8) SAE Feature Alignment (Independent Convergence). We measure the cosine similarity between monosemantic features extracted in an unsupervised manner by a Sparse Autoencoder (SAE) and the semantic directions of Token-Mixture. By testing whether two independent methodologies—differing in training objective, input space, and sparsity enforcement—converge on similar structure, we assess whether the discovered structure reflects the model's internal organization rather than a learning artifact.

Full details of the experimental configuration are provided in Appendix A.

4.4 SPARSE RECOVERY IS SPECIFIC TO THE TOKEN COORDINATE SYSTEM

Whether a semantic direction can be recovered from a small number of tokens depends critically on the coordinate system in which sparsity is imposed. We evaluate this via identity and basis contrasts (Q1, Q2).

Identity contrast. With a budget of $K{=}100$, Token-Mixture recovers 96% of full-vocabulary performance on average (Table 1), whereas Random Token Mixture achieves only 64.2%. The gap is consistent across tasks and is most pronounced for Language. This indicates that semantic signal is concentrated on specific vocabulary elements rather than uniformly distributed across tokens.

Basis contrast. If sparse recovery were merely a consequence of low intrinsic dimensionality, similar performance should be achievable in arbitrary bases. However, replacing the token embedding with a random orthonormal basis drops accuracy to 60–80% at $K{=}100$ for most non-alignment tasks, yielding gaps exceeding 20 percentage points. Sparse recovery therefore holds selectively in the token embedding coordinate system.

Effective rank and cross-model consistency. Effective-rank analysis of the embedding matrix $\mathbf{T}$ (Appendix B) confirms that the representation space is not degenerate at $K{=}100$, ruling out trivial low-rank explanations. Moreover, despite vocabulary sizes ranging from 100K to 201K, all four models maintain over 90% recovery at $K{=}100$, while random bases consistently fail. In small-data settings (e.g., TruthfulQA), the vocabulary-constrained probe occasionally exceeds the unconstrained linear probe, suggesting an implicit regularization effect.

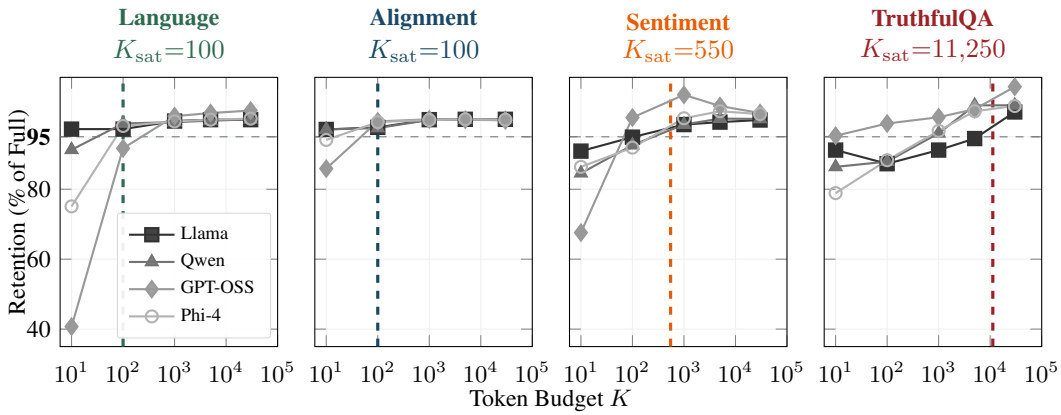

Figure 1: **The number of anchor tokens required to recover a semantic direction reflects the vocabulary-level spread of each concept.** Each panel shows retention curves (percentage of full-vocabulary performance recovered) for four models on a single task. The horizontal dashed line marks the 95% retention threshold. The vertical dashed line marks the task-level saturation point K_{sat} (the smallest K at which the *mean* retention across the four models reaches 95%). Markers denote different models. For each (model, task, K), we evaluate retention at the validation-selected optimal layer.

4.5 DIFFERENT CONCEPTS ACTIVATE DIFFERENT TOKENS

Table 2: Jaccard similarity between tasks within the same model (K=100). Selected tokens show minimal overlap across tasks.

Model	Avg Jaccard	Interpretation
Llama	0.0008	Near zero
Qwen	0.0034	Near zero
GPT-OSS	0.0000	Exactly zero
Phi-4	0.0000	Exactly zero

Token sets across tasks are nearly disjoint. We measure the Jaccard similarity between the active token sets (K=100) of different tasks within the same model. Across all models, the average Jaccard similarity is below 0.004 (Table 2), indicating that each semantic concept activates a distinct region of the vocabulary rather than reusing universally important tokens.

The required number of tokens varies systematically across tasks. The saturation K at which 95% of full-vocabulary performance is reached forms a clear hierarchy (Figure 1):

$$\underset{(K=100)}{\text{Language}} < \underset{(K=100)}{\text{Alignment}} < \underset{(K=550)}{\text{Sentiment}} < \underset{(K=11{,}250)}{\text{TruthfulQA}} \tag{6}$$

This ordering is consistent with the type of signal each task appears to rely on. Language classification primarily draws on localized cues (e.g., script-specific characters and function words) and saturates earliest, whereas truthfulness classification involves more distributed lexical patterns (e.g., entity mentions and discourse-level markers). Below, we provide a qualitative illustration of representative tokens selected for each task.

Qualitative Token Patterns. Across tasks, the selected anchors reflect the structural nature of each signal. Alignment activates policy- and norm-related tokens together with markup fragments, consistent with format-level jailbreak patterns. Truthfulness contrasts template-like fragments with tokens denoting concrete referents. Sentiment combines explicitly evaluative words with topically associated tokens, indicating entanglement beyond pure polarity terms. Language classification is dominated by orthographic markers, confirming that script-level cues alone suffice for early saturation. token lists are provided in Appendix E.3

4.6 THE SELECTED TOKENS ARE CONSTITUTIVE OF THE DIRECTION

In §4.4–4.5, we established that semantic directions can be recovered from a small number of tokens and that the token sets differ across tasks. However, whether these tokens are merely *correlated* with the semantic direction or are actually *constitutive* of it is a separate question. This section addresses this question (Q3) through ablation experiments, steering experiments, and comparison with an independent methodology.

Removing tokens causes the direction to collapse (Causal Necessity). We remove class-specific top tokens from the trained Token-Mixture (without re-training), measure the resulting performance change, and compare against removal of the same number of random tokens. Figure 2 shows the results. Across all tasks, removing class-specific tokens causes

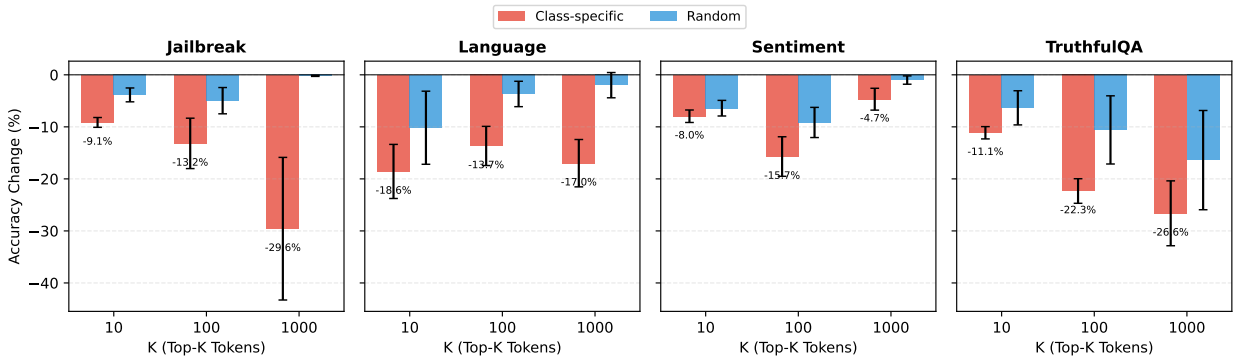

Figure 2: Knock-out ablation results. Removing class-specific top tokens (pink) causes 3–8× greater accuracy degradation than removing random tokens (blue), confirming that Token-Mixture selects semantically essential vocabulary. Results are averaged across all candidate layers for each (model, task) combination.

Table 3: Representative tokens with the highest absolute linear-probe weights for each probing task.

Task	Label	Top-weighted Tokens
Alignment	*Jailbreak*	jailbreak, policy, warning
	Normal	hobby, favorite, beautiful
TruthfulQA	*False*	Protocol, CCA, {{
	Truthful	jurisdiction, directory, Azure
Sentiment	*Negative*	bad, useless, overthrow
	Positive	tender, Friends, marked
Lang. ID	*German*	Über
	Chinese	说
	French	tous
	Spanish	mí
	English	Reflect

3–8× greater accuracy degradation compared to removing random tokens.

The asymmetry has the following implication. Removing tokens without re-training directly tests whether the signal contributed by those tokens can be compensated by other tokens. The fact that random token removal causes only minimal performance degradation reconfirms that the weight of the semantic direction is not uniformly distributed across all tokens but concentrated on a small subset, consistent with the identity contrast in §4.4.

A small number of tokens suffice to steer model output (Causal Sufficiency). If the ablation experiment demonstrated the necessity of the selected tokens, this experiment addresses the converse question: can a sparse combination of tokens achieve steering equivalent to the full dense direction?

We inject into the model's final hidden state $\mathbf{h}$ both the full dense direction $\mathbf{w}_{\text{full}}$ and a sparse direction $\mathbf{w}_{\text{anchor}}$ composed of 200 token embeddings, calibrating the intervention strength λ to achieve a target margin shift.

The output produced by the sparse direction closely matched that of the full direction (mean KL divergence ≈ 0.012, rank preservation 93.9%). In contrast, intervention with a random direction yielded a 3.6× higher KL divergence (0.044). The finding that a token combination comprising only 0.1% of the vocabulary produces a functionally similar effect to a 4096-dimensional dense direction supports the claim that the sparse recovery observed in §4.4 is not merely an approximation.

Convergence with an independent methodology (SAE Feature Alignment). While the causal experiments above demonstrate that "these tokens constitute the direction," comparison with an independent methodology serves to verify that "this structure is not a learning artifact." Sparse Autoencoders (SAEs) extract monosemantic features by decomposing LLM hidden activations in an unsupervised manner [Cunningham et al., 2024], differing from Token-Mixture in training objective, input space, and sparsity enforcement.

We measure the cosine similarity between the 65,536 SAE decoder features at Layer 19 of Llama-3.1-8B-Instruct and the Token-Mixture direction vectors. The average z-score across four tasks is 12.32 ($p < 0.001$), confirming statistically significant alignment relative to the random baseline (Table 4). Jailbreak detection exhibits the highest alignment ($z = 18.74$).

Task	Best Class	Max Align	Top-10 Mean	Z-Score
Jailbreak	jailbreak	0.168	0.142	18.74
TruthfulQA	false	0.141	0.118	13.73
Language	chinese	0.123	0.098	10.38
Sentiment	positive	0.102	0.084	6.44
Average	–	**0.134**	**0.111**	**12.32**

Table 4: Alignment between Token-Mixture semantic directions and SAE monosemantic features. Max Align denotes the maximum cosine similarity among 65,536 SAE features; Z-Score indicates statistical significance relative to the random direction baseline. The mean Max Alignment for the random baseline is 0.068.

Token-Mixture decomposes semantic directions in a supervised manner within the vocabulary embedding coordinate system, whereas SAE extracts features in an unsupervised manner within the hidden activation coordinate system. The fact that two methodologies differing in both objective and space converge on similar structure reduces the likelihood that the observed token-level structure is an artifact of the training procedure.

Summary. Combining the results of the three sections, we arrive at the following picture. Semantic directions are recovered from a small number of tokens in the token embedding coordinate system (§4.4); the token sets are unique to each concept and the required number of tokens reflects the nature of the task (§4.5); and removing the selected tokens causes the direction to collapse, while alignment with an independent methodology is confirmed (§4.6). These results suggest that the vocabulary space may serve not merely as an output interface but as a coordinate system in which semantic directions are organized.

5 DISCUSSION & CONCLUSION

Interpretive implications. The three main findings—(i) sparse recovery specific to the token coordinate system, (ii) near-zero cross-task token overlap, and (iii) causal necessity of selected tokens—converge on a single conclusion: semantic directions are not coordinate-invariant objects in a continuous space but structured entities tied to the token embedding basis. Because LLMs ultimately generate output through vocabulary logits, token embeddings constitute the model's interface with an external symbolic system; the fact that decision boundaries collapse under a random orthonormal basis yet are recovered from $\sim$100 tokens in this basis indicates that the observed sparsity reflects the structure of this interface rather than low-dimensionality of the embedding space (see Appendix B for a formal analysis).

A complementary layer-wise analysis (Appendix D) reveals a U-shaped Token Space Alignment curve—high in early layers, low in middle layers, and high again near the output—mirroring the token-prediction instability observed by Logit Lens [nostalgebraist, 2020] and suggesting a three-phase *lifting–processing–projection* cycle within transformers.

Analytical possibilities opened by token-level decomposition. Shifting the unit of interpretation from "ad-dimensional direction exists" to "which tokens constitute this direction" enables several concrete extensions. *Cross-model comparison*: because token-level decompositions live in a shared vocabulary space rather than model-specific hidden spaces, one can directly examine which anchor tokens are shared across models for the same concept. *Semantic complexity quantification*: the saturation K at which 95% performance is reached (K=100 for language vs. K=11,000 for truthfulness) provides a vocabulary-level measure of concept spread that is complementary to probing accuracy. *Steering transparency*: decomposing a steering vector into readable tokens makes it possible to audit which lexical elements underlie a change in model behavior, a property particularly relevant for safety analysis.

Conclusion. We show that semantic directions in LLMs admit highly sparse representations in the token embedding coordinate system. This sparsity is not a generic property of high-dimensional geometry, but a structural phenomenon specific to the vocabulary basis. Across models and tasks, small anchor sets suffice to reconstruct, differentiate, and causally control semantic decision boundaries. These findings suggest vocabulary space may be interpreted not merely as an output interface, but as a structured coordinate system.

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APPENDIX OVERVIEW

This paper advances four empirically testable claims regarding the structure of semantic directions in the token embedding coordinate system. The appendix is organized so that each claim can be traced to its primary evidence, controlled baselines, and robustness checks. Table 5 provides the complete mapping.

Table 5: Claim–Evidence Map. For each core claim, we list the primary quantitative evidence from the main text, the controlled experiments that rule out alternative explanations, and the robustness analyses that test generality across models, tasks, and layers. Section numbers prefixed with "§" refer to the main text; letters (A–E) refer to appendix sections.

ID	Claim	Primary Evidence	Controls	Robustness
C1	Token coordinate specificity: sparse recovery holds in the token embedding basis but not in arbitrary bases	Table 1 (§4.4): K=100 achieves 96% of full-vocab performance vs. 64% (Rand-Tok) and 77% (Rand-Basis). LinProbe upper bound: 90.7%.	**Identity contrast** (App. A.4): random token selection (RandMix) under identical optimization. **Basis contrast** (App. A.6): random orthonormal basis via QR, 3 seeds. Matched-condition check (Table 10).	4 models $\times$ 4 tasks (Table 1). Effective rank analysis confirms embedding space is not degenerate at K=100 (§4.4). Consistent across vocab sizes 100K–201K.
C2	Concept-specific lexical support: each concept activates a distinct, minimally overlapping token set	Table 2 (§4.5): cross-task Jaccard $\approx$ 0 at K=100. Fig. 1 (§4.5): saturation hierarchy Language (K=100) $<$ Alignment (K=100) $<$ Sentiment (K=550) $<$ TruthfulQA (K=11K).	Cross-task Jaccard protocol: pairwise overlap measured within each model (§4.5).	4 models reproduce the same K_{sat} ordering (Fig. 1). Multi-task TSA comparison (App. D.6, Fig. 7). Minimum-K analysis across layers (App. D.5, Fig. 6).
C3	Causal constitutiveness: anchor tokens are functionally essential, not merely correlated	Fig. 2 (§4.6): class-specific knock-out causes 3–8$\times$ greater degradation than random removal. Steering (§4.6): sparse direction (K=200) yields KL $\approx$ 0.012 vs. full dense direction; random direction KL $=$ 0.044.	**KO control** (App. A.5): random token removal from active set $\mathcal{A}_K$. **Steering control** (§4.6): random direction intervention.	Results averaged across all candidate layers per (model, task) combination (Fig. 2). Layer-wise TSA analysis independently corroborates token–representation coupling (App. D.2, Fig. 3).
C4	Independent convergence: Token-Mixture directions align with unsupervised SAE features	Table 4 (§4.6): mean max cosine alignment $=$ 0.134; mean z-score $=$ 12.32 ($p <$ 0.001) against random baseline ($\mu_{\text{rand}} = 0.068$).	100 random unit vectors as null distribution (App. A.7). SAE and Token-Mixture differ in objective, input space, and sparsity mechanism.	4 tasks $\times$ per-class analysis. Jailbreak shows highest alignment ($z = 18.74$).
C5	Human interpretability: token lists alone support human classification	Fig. 8 (App. E.1): 100% accuracy on Jailbreak, 92% on Sentiment from token lists without original text. Overall interpretability rating: 3.83/5 (Likert).	Participants received only token lists, no original input text.	4 tasks evaluated. Jailbreak: 4.2 ± 1.1; Sentiment: 3.4 ± 1.4; Language: 3.8 ± 1.3; TruthfulQA: 3.2 (Likert).

Appendix organization. The appendix is structured to support efficient verification of the claims above:

- **Appendix A** (Experimental Protocol & Reproducibility) provides full implementation details—compute environment, model specifications, training procedures, baseline implementations, ablation protocols, and a reproducibility checklist—enabling independent replication of all reported results.
- **Appendix B** (Theoretical Conditions) formalizes the structural conditions under which Token-Mixture representation is expected to hold, including sufficient conditions, failure modes, and scope of applicability.
- **Appendix C** (Additional Baselines) reports comparisons with Diff-Means, MLP, and NN-L1 to rule out alternative explanations based on classifier complexity or soft sparsification artifacts.

- **Appendix D** (Token Space Alignment Analysis) presents a layer-wise analysis of how token-space alignment varies across layers, revealing a three-phase processing cycle (token-aligned $\rightarrow$ abstract $\rightarrow$ token-aligned) that is consistent across models and tasks.
- **Appendix E** (Human Interpretability & Additional Results) reports the user study, representative failure cases, and qualitative token lists.

A EXPERIMENTAL PROTOCOL & REPRODUCIBILITY

This appendix provides all implementation details necessary for independent replication of the reported results. We describe the model and representation extraction pipeline (A.1), dataset construction (A.2), Token-Mixture training procedure (A.3), baseline implementations (A.4), knock-out ablation protocol (A.5), random orthonormal basis control (A.6), SAE alignment protocol (A.7), and a reproducibility checklist (A.8).

A.1 MODELS & REPRESENTATION EXTRACTION

Model loading. All models were loaded via HuggingFace's `AutoModelForCausalLM` with `torch_dtype="auto"` and `trust_remote_code=True`. When a model configuration contains a `text_config` attribute, it is used preferentially to ensure compatibility with multimodal architectures. Table 6 summarizes the architectural specifications of the four LLMs used in this work.

Table 6: Architectural specifications of the four LLMs used in all experiments.

Model	Hidden dim. (D)	Layers	Vocab size (V)
Llama-3.1-8B-Instruct	4,096	32	128,256
Qwen2.5-7B-Instruct	3,584	28	152,064
GPT-OSS-20B	6,144	24	201,088
Phi-4	5,120	40	100,352

Input formatting (chat template). To maintain consistent input formatting across models, we applied each model's official chat template to the raw text of every dataset:

```
messages = [{"role": "user", "content": text}]
formatted = tokenizer.apply_chat_template(
    messages, tokenize=False,
    add_generation_prompt=False)
```

Hidden state extraction (last-token pooling). In autoregressive models, the hidden state at the last token position aggregates contextual information from the entire input sequence. For an input sequence of length L, we extract the hidden state at layer l and the last token position: $\mathbf{h}_l(\mathbf{x}) = \mathbf{h}_l^{(L)} \in \mathbb{R}^D$. Tokenization was performed with a maximum sequence length of 512 (`padding=True`, `truncation=True`). The feature extraction batch size was set to 8.

Layer selection (candidate layer search). For each (model, task, K) combination, Token-Mixture and all baselines follow the same layer-selection rule: candidate layers are chosen at intervals of 5 starting from layer 5, with the final layer appended. The layer achieving the highest validation accuracy is selected as the representative layer. Table 7 lists the candidate layers for each model.

Table 7: Candidate layers for each model. All methods independently select the optimal layer using only the validation set.

Model	Candidate Layers
Llama-3.1-8B (32 layers)	[5, 10, 15, 20, 25, 30, 31]
Qwen2.5-7B (28 layers)	[5, 10, 15, 20, 25, 27]
GPT-OSS-20B (24 layers)	[5, 10, 15, 20, 23]
Phi-4 (40 layers)	[5, 10, 15, 20, 25, 30, 35, 39]

A.2 DATASETS

Table 8 summarizes the dataset configuration for all four tasks. Train/validation splits are fixed across all experiments, and all datasets are class-balanced unless otherwise noted.

Table 8: Dataset configuration for each classification task.

Task	Classes	Train	Validation	Source
Language (XNLI)	5	25,000	25,000	XNLI
Sentiment (IMDB)	2	25,000	25,000	IMDB
Alignment (Jailbreak)	2	25,000	5,000	Custom
TruthfulQA	2	1,361	273	TruthfulQA

Language classification (XNLI). We use premise sentences from the XNLI dataset Conneau et al. [2018] and reformulate them as a 5-class language classification task (English, Spanish, Chinese, French, German). We sample 5,000 sentences per language for a balanced split of 25,000 train and 25,000 validation examples.

Sentiment classification (IMDB). We use the Stanford IMDB review dataset Maas et al. [2011], loaded from `stanfordnlp/imdb` on HuggingFace. The official test split is used as the validation set.

Alignment classification (Jailbreak). Benign prompts are drawn from OpenHermes-2.5-Filtered, and jailbreak prompts from JailBreakV-28K Luo et al. [2024]. The resulting binary classification dataset contains 25,000 training and 5,000 validation examples, balanced between the two classes.

Truthfulness classification (TruthfulQA). We construct a binary truthful/false classification task from TruthfulQA Lin et al. [2022]. Due to the relatively small dataset size (1,361 training examples), we increase the number of training epochs for this task (see Table 9).

A.3 TOKEN-MIXTURE TRAINING PROTOCOL

Learnable parameters. The Token-Mixture probe uses the LM head matrix (token embedding matrix) $\mathbf{T} \in \mathbb{R}^{V \times D}$ as a frozen basis. The learnable parameters are the mixture weights $\boldsymbol{\beta} \in \mathbb{R}^{V \times C}$ and bias $\mathbf{b} \in \mathbb{R}^{C}$.

Non-negativity constraint (constructive support). The non-negativity constraint $\boldsymbol{\alpha} \geq 0$ is implemented by applying a ReLU activation to the learnable parameters:

$$\boldsymbol{\alpha} = \mathrm{ReLU}(\boldsymbol{\beta}), \quad \boldsymbol{\beta} \in \mathbb{R}^{V \times C}. \tag{7}$$

This ensures all token contributions are non-negative while enabling gradient-based optimization.

Effective weights and logits. Classification is performed as:

$$\mathbf{W} = \mathbf{T}^{\top} \boldsymbol{\alpha}, \quad \mathrm{logits}(\mathbf{x}) = \mathbf{h}(\mathbf{x})^{\top} \mathbf{W} + \mathbf{b}, \tag{8}$$

where $\mathbf{W} \in \mathbb{R}^{D \times C}$ is the effective weight matrix.

Top-K selection and warmup–exploit schedule. The active token set $\mathcal{A}_K$ is updated at the beginning of each epoch based on the cumulative contribution score:

$$s_i = \sum_c \tilde{\alpha}_{i,c}, \quad \tilde{\boldsymbol{\alpha}} = \mathrm{ReLU}(\boldsymbol{\beta}).$$

We employ a warmup–exploit schedule. In the first epoch (exploration phase), all tokens receive gradients to enable initial contribution estimation, with a weak L_1 regularization ($\lambda = 10^{-4}$) to encourage sparsity. From the second epoch onward (exploitation phase), only the top-K tokens by s_i are retained in $\mathcal{A}_K$; all remaining tokens are masked to zero.

Initialization. $\boldsymbol{\beta}$ is initialized from $\mathcal{N}(0, 0.01)$ and the bias $\mathbf{b}$ is initialized to zero.

Optimization. We use AdamW with learning rate 10^{-3} and weight decay 0.01. The loss function is cross-entropy. The training batch size is 128 and the feature extraction batch size is 8.

Table 9 summarizes all hyperparameters. The only task-specific difference is TruthfulQA, for which the number of epochs is increased to 10 due to its smaller training set.

Table 9: Training and feature extraction hyperparameters for each task.

Parameter	IMDB	Jailbreak	TruthfulQA	Language
Learning rate	10^{-3}	10^{-3}	10^{-3}	10^{-3}
Epochs	3	3	10	3
Weight decay	0.01	0.01	0.01	0.01
Optimizer	AdamW			
Loss function	CrossEntropyLoss			
Training batch size	128	128	128	128
Feature extraction batch size	8	8	8	8
Max sequence length	512	512	512	512
Warmup epochs	1	1	1	1
λ_{L_1}	10^{-4}	10^{-4}	10^{-4}	10^{-4}
Initialization	$\mathcal{N}(0, 0.01)$			

Training pseudocode. Algorithm 1 provides the complete training procedure.

Algorithm 1: Token-Mixture Probe Training

Input: Hidden features $\mathbf{H} \in \mathbb{R}^{N \times D}$, labels $\mathbf{Y}$, token embeddings $\mathbf{T} \in \mathbb{R}^{V \times D}$, budget K, epochs E, L_1 coefficient λ
Output: Active token set $\mathcal{A}_K$, weights $\boldsymbol{\alpha}$, bias $\mathbf{b}$
Initialize: $\boldsymbol{\beta} \sim \mathcal{N}(0,\, 0.01)^{V \times C}$, ;
$\mathbf{b} \leftarrow \mathbf{0}^C$

; ▷ **Phase 1: Exploration** (all V tokens receive gradients)
for each mini-batch $(\mathbf{H}_b, \mathbf{Y}_b)$ in epoch 1 **do**
  $\boldsymbol{\alpha} \leftarrow \mathrm{ReLU}(\boldsymbol{\beta})$;
  $\mathbf{W} \leftarrow \mathbf{T}^\top \boldsymbol{\alpha}$;
  $\mathcal{L} \leftarrow \mathrm{CE}\big(\mathbf{H}_b^\top \mathbf{W} + \mathbf{b},\ \mathbf{Y}_b\big) + \lambda \|\boldsymbol{\alpha}\|_1$;
  Update $\boldsymbol{\beta}, \mathbf{b}$ via AdamW on $\mathcal{L}$;

; ▷ **Phase 2: Exploitation** (top-K mask applied)
$s_i \leftarrow \sum_c \mathrm{ReLU}(\beta_{i,c})$ for all $i \in \{1, \ldots, V\}$;
$\mathcal{A}_K \leftarrow$ top-K tokens ranked by s_i ;
$\beta_{i,:} \leftarrow 0$ for all $i \notin \mathcal{A}_K$; ▷ mask inactive tokens
for each mini-batch $(\mathbf{H}_b, \mathbf{Y}_b)$ in epochs $2, \ldots, E$ **do**
  $\boldsymbol{\alpha} \leftarrow \mathrm{ReLU}(\boldsymbol{\beta})$; ▷ only $\mathcal{A}_K$ entries nonzero
  $\mathbf{W} \leftarrow \mathbf{T}^\top \boldsymbol{\alpha}$;
  $\mathcal{L} \leftarrow \mathrm{CE}\big(\mathbf{H}_b^\top \mathbf{W} + \mathbf{b},\ \mathbf{Y}_b\big)$;
  Update $\boldsymbol{\beta}, \mathbf{b}$ via AdamW on $\mathcal{L}$;
return $\mathcal{A}_K$, $\boldsymbol{\alpha} = \mathrm{ReLU}(\boldsymbol{\beta})$, $\mathbf{b}$

A.4 BASELINE IMPLEMENTATIONS

All baselines use the same representation extraction pipeline (chat template, last-token pooling) and are evaluated on the same train/validation splits as Token-Mixture. Unless otherwise noted, all baselines share the same optimization settings (AdamW, lr $= 10^{-3}$, weight decay $= 0.01$, epochs $= 3$).

Difference-in-Means (Diff-Means). A non-parametric baseline. For each class c, the centroid of the training representations is computed: $\boldsymbol{\mu}_c = \frac{1}{|\mathcal{X}_c|} \sum_{\mathbf{x} \in \mathcal{X}_c} \mathbf{h}(\mathbf{x})$. At inference, the class with the highest cosine similarity to the input is assigned:

$$\hat{y} = \arg\max_c \frac{\mathbf{h}(\mathbf{x}) \cdot \boldsymbol{\mu}_c}{\|\mathbf{h}(\mathbf{x})\|\, \|\boldsymbol{\mu}_c\|}. \tag{9}$$

No learnable parameters are involved.

Linear Probe (LinProbe). A standard linear classifier $\mathbf{W} \in \mathbb{R}^{D \times C}$ trained with cross-entropy loss. No constraint is imposed on the weight matrix. Architecture: `nn.Linear(`D`, `C`)`.

MLP. A two-layer MLP with hidden dimension 512, ReLU activation, and dropout 0.1. Architecture: `Linear(`D`, 512)` $\rightarrow$ `ReLU` $\rightarrow$ `Dropout(0.1)` $\rightarrow$ `Linear(512, `C`)`. Trained with the same optimization settings as Token-Mixture.

Random Token Mixture (RandMix). This baseline tests whether Token-Mixture's performance arises from *meaningful token selection* or from a mere dimensionality reduction effect. Instead of selecting the top-K tokens by contribution score, K tokens are selected uniformly at random: `np.random.RandomState(seed).choice(`V`, `K`, replace=False)`. All other training procedures (non-negativity constraint, optimization settings, hyperparameters) are identical to Token-Mixture.

Nonnegative L_1 (NN-L_1). A linear classifier with non-negativity constraint and L_1 regularization ($\lambda = 10^{-3}$), operating in the token similarity space $\mathbf{Th}(\mathbf{x}) \in \mathbb{R}^V$. Unlike Token-Mixture, this baseline does not apply a hard top-K mask; instead, it learns weights over the full vocabulary and induces soft sparsity via L_1:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \sum_{i=1}^{V} \sum_{c=1}^{C} \text{ReLU}(\beta_{i,c}). \tag{10}$$

After training, the number of active tokens is computed using a threshold of 10^{-4}: a token i is considered active if $\sum_c \text{ReLU}(\beta_{i,c}) > 10^{-4}$. This baseline isolates the effect of the explicit top-K budget from the structural properties of the token embedding coordinate system.

A.5 KNOCK-OUT ABLATION PROTOCOL

To verify that Token-Mixture depends constitutively on specific anchor tokens, we perform knock-out experiments that remove important tokens from a trained probe and measure the resulting performance change.

Removal method (no re-training). Given a fully trained Token-Mixture probe, we set the LM head rows corresponding to the target tokens to zero. The learned parameters ($\boldsymbol{\beta}$, $\mathbf{b}$) are not modified, and no re-training is performed. The modified probe is evaluated directly on the validation set. This setup directly measures whether the decision boundary collapses when specific token contributions are removed.

Removal targets. For each class c, tokens are sorted in descending order by their contribution weight $\alpha_{i,c}$, and the top-N tokens are removed (*class-specific removal*). As a control, the same number N of tokens is selected uniformly at random from the active set $\mathcal{A}_K$ and removed (*random removal*).

Evaluation metrics. We measure overall accuracy, per-class recall, and per-class precision. The key metric is the recall change for the class whose tokens were removed, as this directly reflects whether the removed tokens are essential for that class's decision boundary.

A.6 RANDOM ORTHONORMAL BASIS CONTROL

This control experiment tests whether the high performance of Token-Mixture arises from the semantic structure of the token embedding basis or merely from sparse projection in a low-dimensional space.

Experimental design. We replace only the basis while keeping all other components of Token-Mixture identical. Instead of the selected token embeddings $\mathbf{T}_{\text{selected}} \in \mathbb{R}^{K \times D}$, we use a random orthonormal basis $\mathbf{B} \in \mathbb{R}^{K \times D}$ generated via QR decomposition.

Orthonormal basis generation (QR). K orthonormal vectors are generated as follows (reduced QR for $K \leq D$):

1. Sample a Gaussian matrix $\mathbf{G} \in \mathbb{R}^{D \times K}$.
2. Compute the QR decomposition $\mathbf{G} = \mathbf{QR}$ to obtain $\mathbf{Q} \in \mathbb{R}^{D \times K}$ (orthonormal columns).

3. Use $\mathbf{B} = \mathbf{Q}^\top \in \mathbb{R}^{K \times D}$ as the basis.

Training conditions. The basis $\mathbf{B}$ is frozen, and only the learnable parameters $\boldsymbol{\alpha} \in \mathbb{R}^{K \times C}$ and bias $\mathbf{b} \in \mathbb{R}^C$ are trained. All other conditions—non-negativity constraint (ReLU), learning rate (10^{-3}), weight decay (0.01), batch size (128)—are identical to Token-Mixture.

Fairness guarantee. Table 10 compares the experimental conditions between the token basis and the random basis. Critically, the random basis satisfies unit norm and perfect orthogonality, which are properties *not* guaranteed by the token basis. The random basis is thus *more favorable* in terms of numerical conditioning. If the random basis nonetheless yields lower performance, this provides stronger evidence for the structural specificity of the token embedding coordinate system.

Table 10: Experimental condition comparison between the token basis and the random orthonormal basis.

Factor	**Token Basis**	**Random Basis**	**Matched?**
Learnable parameters	$K \times C + C$	$K \times C + C$	✓
Non-negativity constraint	ReLU($\boldsymbol{\beta}$)	ReLU($\boldsymbol{\beta}$)	✓
Optimizer / LR / Epochs	Identical	Identical	✓
Representation extraction	Identical	Identical	✓
Basis normalization	Variable norm	Unit norm	×
Basis orthogonality	Non-orthogonal	Fully orthogonal	×

A.7 SAE ALIGNMENT PROTOCOL

This section describes how we measure the alignment between Token-Mixture semantic directions and monosemantic features extracted by a Sparse Autoencoder (SAE).

SAE model. We use `goodfire/Llama-3.1-8B-Instruct-SAE-l19`, a Sparse Autoencoder trained on the layer 19 hidden states of Llama-3.1-8B-Instruct. The decoder matrix $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{M \times D}$ contains $M = 65{,}536$ SAE features with $D = 4{,}096$.

Token-Mixture direction vectors. For each class c, the Token-Mixture direction vector is constructed from the learned weights:

$$\mathbf{w}_c = \mathbf{T}_{\text{selected}}^\top \boldsymbol{\alpha}_c, \tag{11}$$

where $\boldsymbol{\alpha}_c \in \mathbb{R}^K$ is the token weight vector for class c.

Alignment measurement. We compute the cosine similarity between the Token-Mixture direction $\mathbf{w}_c$ and each SAE feature $\mathbf{w}_{\text{dec},j}$:

$$\text{sim}(j) = \frac{\mathbf{w}_c \cdot \mathbf{w}_{\text{dec},j}}{\|\mathbf{w}_c\| \, \|\mathbf{w}_{\text{dec},j}\|}. \tag{12}$$

We report the cosine similarity of the top-10 SAE features ranked by absolute similarity.

Random baseline and statistical significance. To assess whether the observed alignment exceeds chance, we compute alignment statistics for 100 random direction vectors sampled from $\mathcal{N}(0, 1)$ and L_2-normalized. Statistical significance is assessed via the z-score:

$$z = \frac{\text{max_align} - \mu_{\text{random}}}{\sigma_{\text{random}}}, \tag{13}$$

where $z > 2$ is considered statistically significant.

Scope. The analysis is conducted on Llama-3.1-8B-Instruct at layer 19 with $K = 100$, for all four tasks (Language, Sentiment, TruthfulQA, Jailbreak).

B THEORETICAL CONDITIONS FOR TOKEN-MIXTURE REPRESENTATION

This appendix formalizes the structural conditions under which Token-Mixture enables interpreting semantic directions as sparse combinations of LM head token embeddings. We state the sufficient conditions, provide structural motivations for why token embeddings serve as a meaningful basis, discuss the optimization formulation, and characterize the failure modes.

B.1 SUFFICIENT CONDITIONS FOR TOKEN-BASED SPARSE REPRESENTATION

Token-Mixture holds for autoregressive LLMs when the following three conditions are satisfied.

Condition 1: Next-token prediction objective. Transformer-based language models generally adopt a factorized output form:

$$P(x_{t+1} \mid \mathbf{h}_t) = \text{softmax}(\mathbf{T}\mathbf{h}_t), \tag{14}$$

where $\mathbf{h}_t \in \mathbb{R}^D$ is the hidden state at a given layer and $\mathbf{T} \in \mathbb{R}^{V \times D}$ is the LM head token embedding matrix. This structure implies that the model reads semantic information through linear projections onto token embedding directions.

Condition 2: Hidden states lie approximately within the span of token embeddings. The residual stream in Transformers iteratively combines token embeddings, so hidden states $\mathbf{h}$ can be expressed approximately as:

$$\mathbf{h} \approx \sum_{i=1}^{V} \alpha_i \mathbf{T}_i, \tag{15}$$

meaning that hidden states are approximately representable as superpositions of the token embedding basis. This makes the reconstruction of semantic directions as sparse token-based combinations structurally well-founded.

Condition 3: Linear separability in hidden space. Extensive prior work has demonstrated that diverse semantic tasks—including language classification, sentiment analysis, and factuality detection—are linearly separable in hidden state space Alain and Bengio [2017], Belinkov [2022]. Therefore, constraining semantic directions to the form $\mathbf{w} = \mathbf{T}^\top \boldsymbol{\alpha}$ retains sufficient expressive power for capturing these properties.

These three conditions are satisfied by the models evaluated in this work (Llama, Qwen, GPT-OSS, Phi-4) and are expected to hold broadly for autoregressive Transformer LLMs that employ a linear LM head.

B.2 STRUCTURAL MOTIVATIONS FOR TOKEN-BASED DECOMPOSITION

This subsection provides structural reasons why LM head token embeddings serve as a meaningful basis for decomposing semantic directions.

Reason 1: Token embeddings as the model's output coordinate system. When generating text, the model computes:

$$\text{logit}(i) = \langle \mathbf{T}_i, \mathbf{h} \rangle. \tag{16}$$

Each token embedding $\mathbf{T}_i$ represents the direction associated with token i in the model's output space. The model describes the world through inner products with these directions, making token embeddings a natural coordinate system for interpreting semantic structure.

Reason 2: Residual stream as superposition of token embeddings. The residual stream update mechanism—comprising autoregressive attention, MLP transformations, and skip connections—maintains hidden states as transformed combinations of token embeddings. This means that the principal axes of the representation space are structurally related to the token basis.

Reason 3: Training-induced alignment pressure. Throughout training, LLMs receive pressure via the cross-entropy loss to align hidden states with the LM head. Weight tying (sharing input embeddings with the LM head), when used, further reinforces the alignment between the embedding space and the space in which semantic information is encoded.

B.3 OPTIMIZATION FORMULATION

Token-Mixture projects semantic directions into token space through the following optimization problem:

$$\min_{\boldsymbol{\alpha} \geq 0} \|\mathbf{T}^\top \boldsymbol{\alpha} - \mathbf{w}\|_2^2 + \lambda \|\boldsymbol{\alpha}\|_1. \tag{17}$$

When the conditions outlined above are approximately satisfied, $\mathbf{w}$ lies close to the row space of $\mathbf{T}$, and this problem admits a stable solution. The L_1 penalty and non-negativity constraint jointly encourage sparse, interpretable decompositions.

We note that the goal is not exact reconstruction of $\mathbf{w}$ but rather a regularized projection that retains the discriminative information of the semantic direction while achieving interpretability through sparsity.

B.4 FAILURE MODES

Token-Mixture may fail under the following circumstances:

- **Architectures lacking a linear LM head.** Models that do not compute output logits as $\mathbf{Th}$ (e.g., softmax-free architectures) violate Condition 1, and the vocabulary projection central to Token-Mixture is not defined.
- **Hidden states outside the token embedding span.** If the model's hidden states are not well-approximated by linear combinations of token embeddings (e.g., due to highly nonlinear transformations that push representations far from the token embedding subspace), Condition 2 is violated and the reconstruction quality may degrade.
- **Non-linearly separable tasks.** If a semantic task requires a nonlinear decision boundary in hidden space, constraining the direction to $\mathbf{w} = \mathbf{T}^\top \boldsymbol{\alpha}$ may be insufficient, as this restricts the boundary to a linear form within the token embedding row space.

In practice, these failure conditions are uncommon for the standard autoregressive Transformer architectures evaluated in this work. However, whether Token-Mixture extends to substantially different architectures (e.g., state-space models, encoder-only models, or multimodal architectures) requires further investigation.

B.5 SCOPE OF APPLICABILITY

In summary, Token-Mixture leverages structural properties that are common to modern autoregressive Transformer LLMs:

1. These architectures employ token embedding-based output coordinate systems (Condition 1).
2. Hidden states maintain approximate alignment with the token embedding space (Condition 2).
3. Many semantic tasks of interest are linearly separable in hidden space (Condition 3).

When these conditions are satisfied, Token-Mixture provides a principled framework for decomposing semantic directions into sparse, interpretable token combinations. Its applicability extends to architectures satisfying Conditions 1–3, though the degree of sparsity and reconstruction quality may vary depending on model architecture, vocabulary structure, and task characteristics.

C ADDITIONAL BASELINES & ALTERNATIVE EXPLANATIONS

The primary evidence for Claim C1 (token coordinate specificity) is provided by the identity contrast (Rand-Tok) and basis contrast (Rand-Basis) experiments reported in the main text (§5.1). This appendix supplements those results by reporting additional baselines that serve two purposes: (i) confirming that the proposed method does not sacrifice excessive performance relative to unconstrained classifiers, and (ii) ruling out alternative explanations based on classifier complexity or soft sparsification artifacts.

C.1 COMPARED METHODS

Linear Probe (LinProbe). An unconstrained linear classifier operating on the hidden state $\mathbf{h}(\mathbf{x}) \in \mathbb{R}^D$. This provides an upper bound on linear separability in the representation space, as the weight matrix $\mathbf{W} \in \mathbb{R}^{D \times C}$ is free to span all D dimensions.

MLP. A one-hidden-layer multilayer perceptron with hidden dimension 512, ReLU activation, and dropout 0.1. This tests whether nonlinear decision boundaries substantially improve performance over the linear constraint imposed by Token-Mixture.

Diff-Means. A non-parametric baseline that assigns the class whose centroid has the highest cosine similarity to the input. This represents the weakest baseline: if Diff-Means already achieves high accuracy, then statistical separability alone—without any learned decision boundary—suffices for the task.

NN-L_1 (Soft Sparse Classifier). A linear classifier with non-negativity constraint and L_1 regularization ($\lambda = 10^{-3}$), operating in the token similarity space $\mathbf{Th}(\mathbf{x}) \in \mathbb{R}^V$. Unlike Token-Mixture, NN-L_1 does not apply a hard top-K mask; it learns weights over the full vocabulary and induces soft sparsity via L_1 regularization. This baseline isolates whether the sparse recovery phenomenon arises from the specific top-K selection strategy or from the structural properties of the token embedding coordinate system itself.

Full implementation details for all baselines are provided in Appendix A.4.

C.2 RESULTS

Table 11 presents the unified comparison. Token-Mixture results at varying token budgets ($K = 10$ to Full) are shown alongside all baselines. The "Eff." column reports the efficiency ratio: Token-Mixture accuracy at K=100 as a percentage of the full-vocabulary accuracy.

C.3 INTERPRETATION

Three main observations emerge from this comparison.

Token-Mixture is not excessively constrained. LinProbe and MLP generally achieve slightly higher performance than Token-Mixture, which is expected given that they operate without vocabulary or sparsity constraints. The average gap between LinProbe (87.4%) and Token-Mixture at K=100 (86.5%) is less than 1 percentage point, indicating that the vocabulary row-space constraint does not substantially sacrifice discriminative power. Notably, for TruthfulQA on Llama, Token-Mixture with the full vocabulary (86.5%) exceeds LinProbe (77.0%), suggesting that the vocabulary constraint can act as an implicit regularizer that prevents overfitting on small datasets.

Sparse recovery is not an artifact of a specific sparsification technique. NN-L_1 achieves moderate performance (74.0% average) but does not reproduce the same recovery characteristics as Token-Mixture at comparable sparsity levels. Specifically, NN-L_1 induces soft sparsity via L_1 regularization without an explicit token budget, yet it falls substantially short of Token-Mixture at K=100 (86.5%). This indicates that the structured sparse recovery observed in Token-Mixture is not a byproduct of any particular sparsification trick, but arises from the interaction between the hard top-K selection and the semantic structure of the token embedding coordinate system.

Table 11: Unified baseline comparison. Left block: baseline methods (Diff-Means, LinProbe, RandMix, MLP, NN-L_1). Right block: Token-Mixture performance as a function of token budget K. Eff. denotes the ratio of K=100 performance to full-vocabulary performance. For each (model, task) combination, the optimal layer is independently selected based on validation accuracy.

		Baselines					**Token-Mixture (K tokens)**					
Task	**Model**	Diff	LinP	RandM	MLP	NN-L_1	10	100	1K	5K	Full	Eff.
Language	Qwen	63.1	98.6	46.9	98.3	74.2	91.2	98.6	99.0	99.4	99.8	99%
	Llama	82.7	98.7	50.9	99.0	69.6	97.0	97.0	99.3	99.7	99.8	97%
	GPT-OSS	30.3	82.4	43.8	90.0	49.7	37.9	85.6	94.2	95.0	93.3	92%
	Phi-4	51.2	96.3	46.8	95.3	86.1	74.7	97.6	98.8	99.4	99.5	98%
Alignment	Qwen	94.0	99.7	78.9	99.5	98.1	96.9	97.3	99.5	99.7	99.7	98%
	Llama	95.8	99.9	91.9	100.0	97.6	97.1	97.7	99.8	100.0	100.0	98%
	GPT-OSS	68.1	99.6	89.3	99.4	98.5	85.6	99.1	99.6	99.5	99.6	99%
	Phi-4	92.5	99.8	94.1	99.8	98.7	94.0	99.1	99.8	99.9	99.9	99%
Truthful	Qwen	59.6	71.7	52.9	72.2	53.2	63.0	64.1	70.0	75.8	72.9	88%
	Llama	71.8	77.0	61.5	77.5	69.3	78.8	75.5	78.8	81.7	86.5	87%
	GPT-OSS	56.3	55.3	51.9	56.0	50.0	60.1	62.3	63.4	64.8	63.0	99%
	Phi-4	59.3	72.3	57.6	69.6	62.4	61.5	68.9	75.5	79.9	78.0	88%
Sentiment	Qwen	70.4	86.3	60.2	86.9	61.6	75.8	83.0	88.3	89.8	89.6	93%
	Llama	77.6	90.4	71.8	91.5	80.4	85.2	89.0	92.3	92.9	93.7	95%
	GPT-OSS	52.2	80.9	61.2	72.3	66.3	54.3	80.8	86.0	83.4	80.4	100%
	Phi-4	67.9	89.4	67.4	89.9	78.7	78.3	83.3	90.9	92.8	90.7	92%
Average		68.3	87.4	64.2	87.3	74.0	77.0	86.5	89.2	91.3	91.1	96%

Simple centroid separation is insufficient. Diff-Means achieves the lowest average performance (68.3%), demonstrating that naive statistical separability—without a learned decision boundary—does not adequately capture the structure of semantic directions. The gap is most pronounced for Language classification on GPT-OSS (30.3% vs. 85.6% for Token-Mixture at K=100) and for Alignment on GPT-OSS (68.1% vs. 99.1%).

Summary. These additional baselines reinforce the main-text findings by confirming that: (i) Token-Mixture's performance is not explained by classifier complexity, as unconstrained classifiers yield only marginal gains; (ii) the sparse recovery phenomenon is not reproduced by soft sparsification alone; and (iii) simple centroid-based separation is insufficient. Combined with the coordinate system controls (Rand-Tok, Rand-Basis) in §5.1, these results support the interpretation that sparse recovery of semantic directions is a structural property of the token embedding coordinate system, rather than an artifact of the training procedure or sparsification strategy.

D TOKEN SPACE ALIGNMENT ANALYSIS

This appendix presents a layer-wise analysis of how well hidden representations align with the vocabulary token space. By normalizing Token-Mixture probe accuracy into the Token Space Alignment (TSA) metric, we quantify, for each layer, what fraction of classification information can be captured using only K token basis vectors. Five complementary analyses—TSA curves, cross-layer token similarity, token weight diversity, minimum-K thresholds, and multi-task TSA comparison—converge on a unified interpretation: transformers undergo a three-phase processing cycle in which representations transition from the token basis to an abstract basis and back.

D.1 ANALYSIS FRAMEWORK

Definition. We define the Token Space Alignment at layer l with token budget K as:

$$\mathrm{TSA}(l, K) = \frac{\mathrm{Acc}_{\mathrm{TM}}(l, K)}{\mathrm{Acc}_{\mathrm{TM}}(l, K_{\mathrm{full}})}, \tag{18}$$

where $\mathrm{Acc}_{\mathrm{TM}}(l, K)$ is the validation accuracy of a Token-Mixture probe using only the top-K tokens as the basis at layer l, and K_{full} is the full vocabulary size (e.g., 128,256 for Llama-3.1-8B). Since $\mathrm{Acc}_{\mathrm{TM}}(l, K_{\mathrm{full}})$ closely matches the performance of an unconstrained linear probe, TSA can be interpreted as the fraction of classification information that is captured by K token basis vectors.

Interpretation. TSA ≈ 1 indicates that the representation at that layer is well-aligned with the vocabulary token space: a sparse combination of token embeddings suffices for classification. TSA $\ll 1$ indicates that the representation has moved away from the token space into an abstract processing space where token-basis constraints incur substantial information loss.

Role of K. K functions as a *token-space constraint strength* parameter. As $K \to K_{\mathrm{full}}$, the constraint vanishes (TSA $\to 1$). As K decreases, the constraint tightens and differences in token-space alignment across layers are amplified.

D.2 TSA CURVES

Figure 3 shows the TSA curves for four models across five values of K. The most prominent pattern is a **U-shaped curve**: high TSA in early layers $\to$ a sharp drop in middle layers $\to$ recovery in later layers, forming a three-phase processing cycle.

Phase I: Token-aligned stage (early layers). At Layer 1 of Llama-3.1-8B, $\mathrm{TSA}(1, 100) = 0.981$: 100 tokens suffice to recover 98.1% of the full-vocabulary probe performance. Phi-4 (0.981) and Qwen2.5-7B (0.996 at Layer 3) similarly exhibit TSA > 0.95 in early layers. This indicates that representations produced by the first transformer blocks are strongly aligned with the vocabulary token space. At this stage, classification information is directly encoded in the embedding directions of a small number of language-specific tokens (characters, morphemes, function words).

Phase II: Abstract processing stage (middle layers). At Llama Layer 11, $\mathrm{TSA}(11, 100) = 0.654$, a drop of 33.4 percentage points from the early layers. Phi-4 drops to 0.564 at Layer 16, a loss of 42.5 percentage points. At $K = 10$, the decline is even more dramatic: Llama Layer 8 reaches $\mathrm{TSA}(8, 10) = 0.350$, meaning only 35% of the classification information is captured by 10 tokens.

The interpretation is clear: in middle layers, representations move away from the vocabulary token space into an abstract processing space. The model performs higher-order operations at this stage—contextual meaning integration, long-range dependency processing—that transcend token-level surface patterns. These abstract representations are not aligned with any specific token embeddings, resulting in information loss under the token-basis constraint.

Phase III: Token-return stage (later layers). At Llama Layers 27–29, TSA recovers to approximately 0.93; Phi-4 recovers to 0.88 at Layers 33–39. This recovery reflects the fact that representations near the output must be re-projected into the vocabulary token space for next-token prediction. The LM head operates in the token embedding space, so the final layers must transform abstract representations back into token-aligned form.

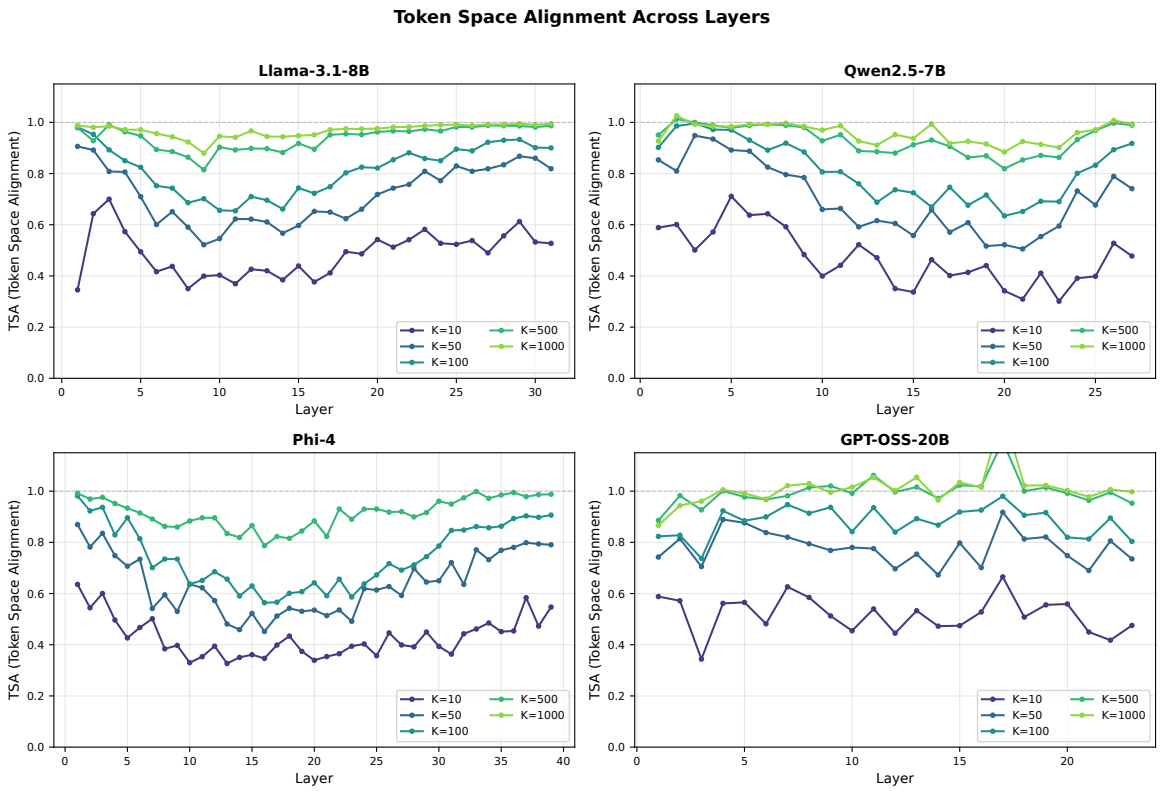

Figure 3: Layer-wise TSA curves for four models. Smaller K produces a more pronounced U-shaped pattern, reflecting the three-phase processing cycle in which representations depart from the vocabulary token space in middle layers and return near the output.

Quantitative summary. The U-shape score, defined as $\frac{\mu_{\text{early}}+\mu_{\text{late}}}{2} - \mu_{\text{middle}}$, is $+0.274$ for Phi-4 at $K = 100$ (the strongest), $+0.165$ for Llama and Qwen, and -0.047 for GPT-OSS. The exception for GPT-OSS (absence of a clear U-shape) is attributed to its relatively shallow architecture (24 layers), which compresses the abstract processing stage.

Symmetry with Logit Lens. Logit Lens nostalgebraist [2020] projects middle-layer hidden states through the LM head to observe which tokens the model "thinks about." TSA operates in precisely the reverse direction: it constrains classification to the token embedding basis and tracks performance changes. Both perspectives capture the same phenomenon—departure from the token space in middle layers and return near the output—confirming that the three-phase cycle is a fundamental structural property of transformer processing, observable from both forward (hidden state $\rightarrow$ vocabulary) and reverse (vocabulary $\rightarrow$ classification) directions.

Effect of K: a continuum of token-space constraint. At $K = 1{,}000$, the U-shape score drops to $+0.022$ for Llama and $+0.024$ for Qwen, nearly vanishing. This means that 1,000 tokens can capture most classification information even in the abstract processing stage, indicating that "abstract" representations are not fully orthogonal to the token space but rather require finer-grained combinations of more tokens. K thus functions as a *resolution parameter* for measuring token-space alignment: smaller K applies a stricter criterion.

D.3 CROSS-LAYER TOKEN SIMILARITY

Figure 4 visualizes the weighted cosine similarity between the top-100 tokens selected at each pair of layers within the same model. Each layer is represented as a sparse vector $\{\text{token_id} : \text{weight}\}$, and the inter-layer cosine similarity measures the degree to which two layers utilize the same tokens with similar weights.

A **block-diagonal structure** is observed across all models:

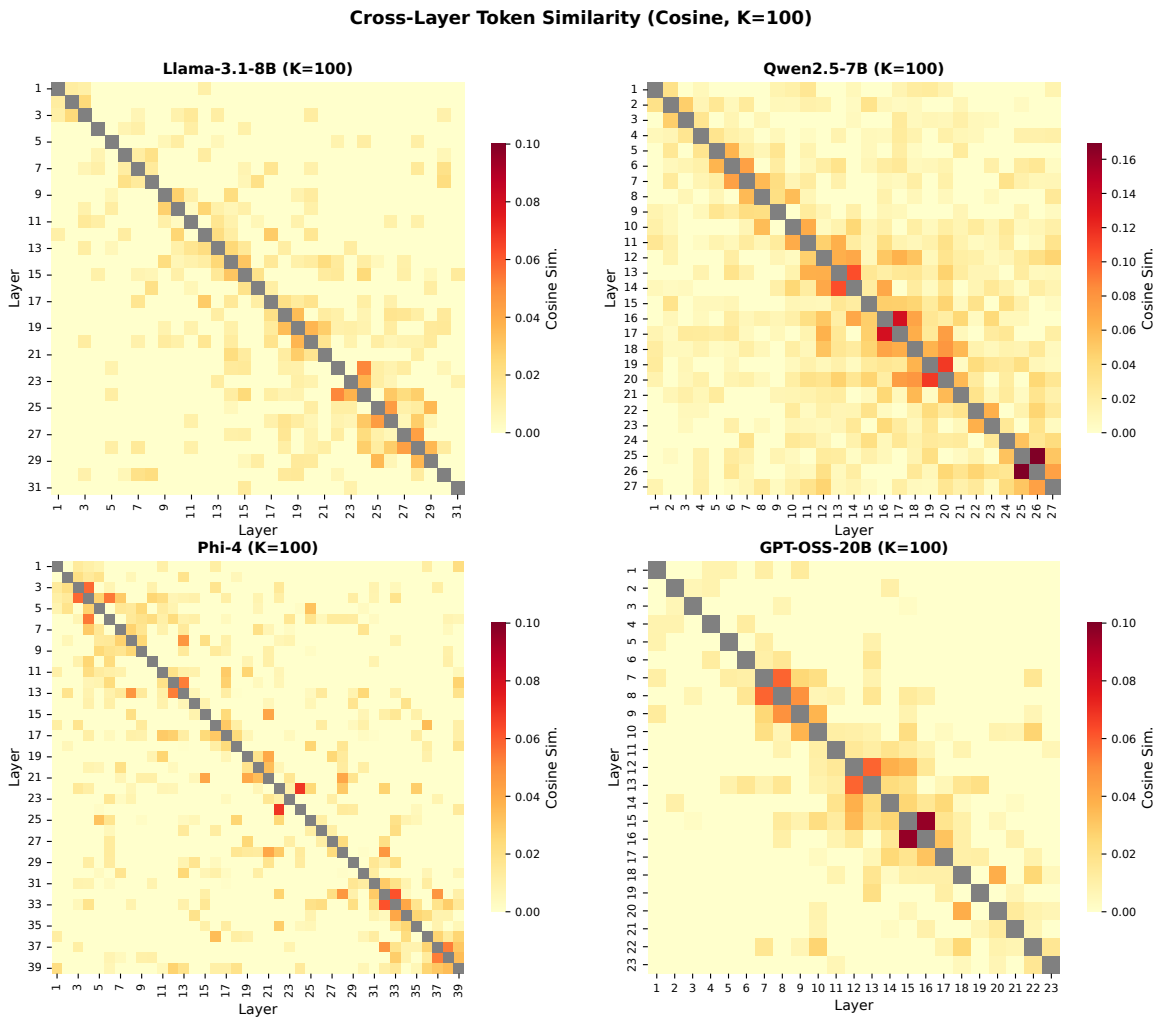

Figure 4: Cosine similarity heatmaps of selected tokens ($K = 100$) between all layer pairs. Diagonal entries (self-similarity) are masked in gray. High within-block similarity in early/late layers and low inter-block similarity confirm the three-phase processing cycle at the spatial level.

Early block (Llama L1–7, Qwen L1–7, Phi-4 L1–7, GPT-OSS L1–5): High similarity between adjacent layers. These layers share similar surface-level tokens (characters, morphemes), reflecting the token-aligned processing of Phase I.

Middle gap (Llama L8–20, Phi-4 L10–25): Inter-block similarity drops sharply, approaching zero. Middle layers either use distinct abstract token sets or form distributed representations that are not strongly aligned with any specific tokens.

Late block (Llama L25–31, Qwen L24–27, Phi-4 L33–39, GPT-OSS L15–23): High within-block similarity re-emerges. Layers near the output converge on similar token sets, spatially corroborating the token-return phase (Phase III).

A notable exception is Qwen2.5-7B, where an additional hotspot appears at Layers 15–20, suggesting that the Qwen architecture partially maintains token-space alignment even in middle layers.

Interpretation. The block-diagonal structure corroborates the U-shaped TSA curves at the *layer-pair level*. Early and late layers each share a consistent set of "token-space-aligned" tokens, while middle layers depart from this set. This is consistent with a *basis transition* process: the transformer begins in the vocabulary basis, transitions to an abstract basis, and returns to the vocabulary basis for output generation.

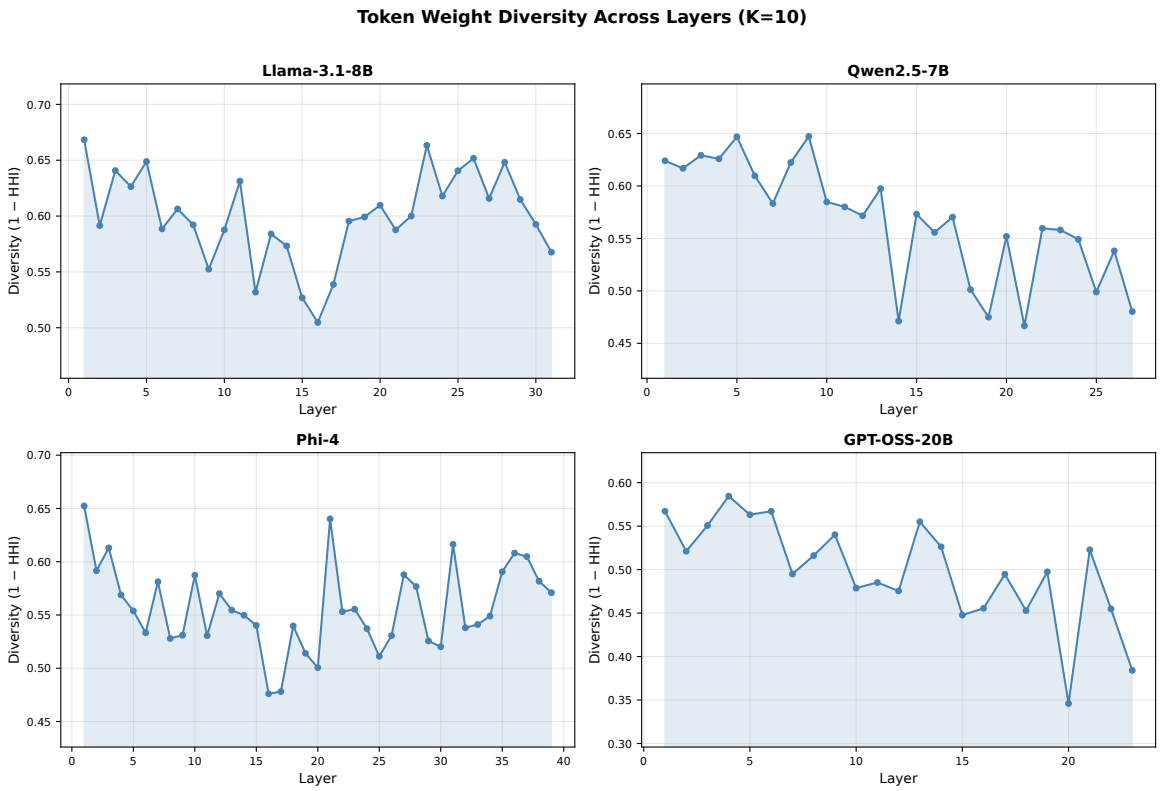

Figure 5: Layer-wise token weight diversity ($1 - \text{HHI}$) at $K = 10$. Low diversity indicates tokens specialized for a single language; high diversity indicates general-purpose tokens whose weights are distributed across languages.

D.4 TOKEN WEIGHT DIVERSITY

We measure the concentration of each token's weight distribution across the five language classes using the Herfindahl–Hirschman Index (HHI), and define diversity as:

$$D_i = 1 - \sum_{c=1}^{C} \left(\frac{w_{i,c}}{\sum_{c'} w_{i,c'}} \right)^2, \tag{19}$$

where $w_{i,c}$ is the weight of token i for class c. $D \approx 0$ indicates a token specialized for a single language (e.g., a Chinese-specific character token $\rightarrow$ Chinese), while $D \approx 0.8$ indicates a general-purpose token whose weight is distributed evenly across all five languages.

Figure 5 shows the layer-wise mean diversity at $K = 10$.

Llama-3.1-8B. Diversity drops from $D = 0.67$ at Layer 1 to $D = 0.50$ at Layer 16 (a 25% decline), then recovers to $D = 0.65$ at Layer 25. In early layers, the 10 selected tokens are general-purpose tokens shared across languages (e.g., code tokens, special symbols). In middle layers, each token becomes specialized for a specific language.

This presents an apparent paradox: TSA is *low* in middle layers, yet the few selected tokens are *more* language-specific. The resolution is that classification information in middle layers is concentrated on a small number of specialized tokens, but these tokens alone are insufficient to capture all the information (hence the low TSA). The information is spread across the token space, making it difficult to capture with the narrow basis of $K = 10$.

GPT-OSS-20B. This model shows a distinctive monotonically decreasing diversity pattern ($0.58 \rightarrow 0.35$), meaning that later layers select increasingly language-specialized tokens. This is consistent with GPT-OSS's lack of a clear U-shaped TSA pattern: the shallow 24-layer architecture does not fully deploy the "abstract $\rightarrow$ return" cycle but instead undergoes gradual specialization.

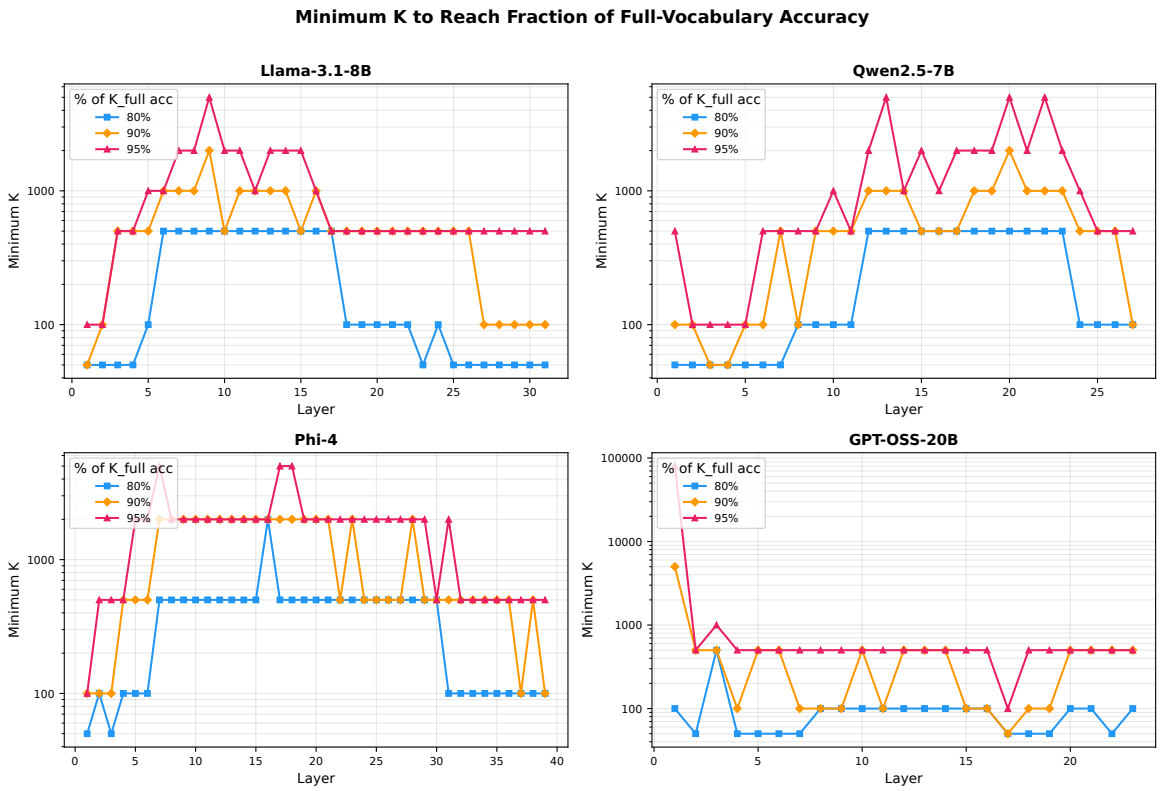

Figure 6: Minimum K required to reach 80%/90%/95% of full-vocabulary accuracy at each layer (log scale). The inverted U-shaped pattern shows that middle layers require substantially more tokens, reflecting the departure from token-space alignment.

D.5 MINIMUM-K THRESHOLD—LAYER-WISE INFORMATION COMPRESSION

Figure 6 shows, for each layer, the minimum K required to achieve $\text{TSA}(l, K) \geq \theta$ for $\theta \in \{0.80, 0.90, 0.95\}$. This metric directly measures how compressed the classification information is over the token space at each layer.

An **inverted U-shaped** pattern is observed across all models:

Early layers (L1–3): $K = 10$–50 suffices to reach the 80% threshold. Classification information is captured by as few as 10–50 tokens, indicating extreme compression in the token space.

Middle layers (L8–20): $K = 500$–$2{,}000$ is required. At Phi-4 Layers 16–18, the 95% threshold demands $K = 5{,}000$—over $100\times$ more tokens than the early layers.

Late layers (L25+): Minimum K decreases back to 50–100.

Interpretation. In early layers, classification information is concentrated in the embedding directions of a small number of "representative tokens" (e.g., high-frequency Chinese function words and English tokens such as "the" and "is"). In middle layers, representations are abstracted beyond these surface-level token patterns, requiring far more tokens in fine-grained weighted combinations to capture equivalent information. The decrease in minimum K in later layers confirms that information is re-compressed into a small number of core token directions as representations are re-aligned for output.

This inverted U-shape reinterprets the TSA curves (Analysis 1) from an "information compression" perspective, independently confirming the same three-phase cycle.

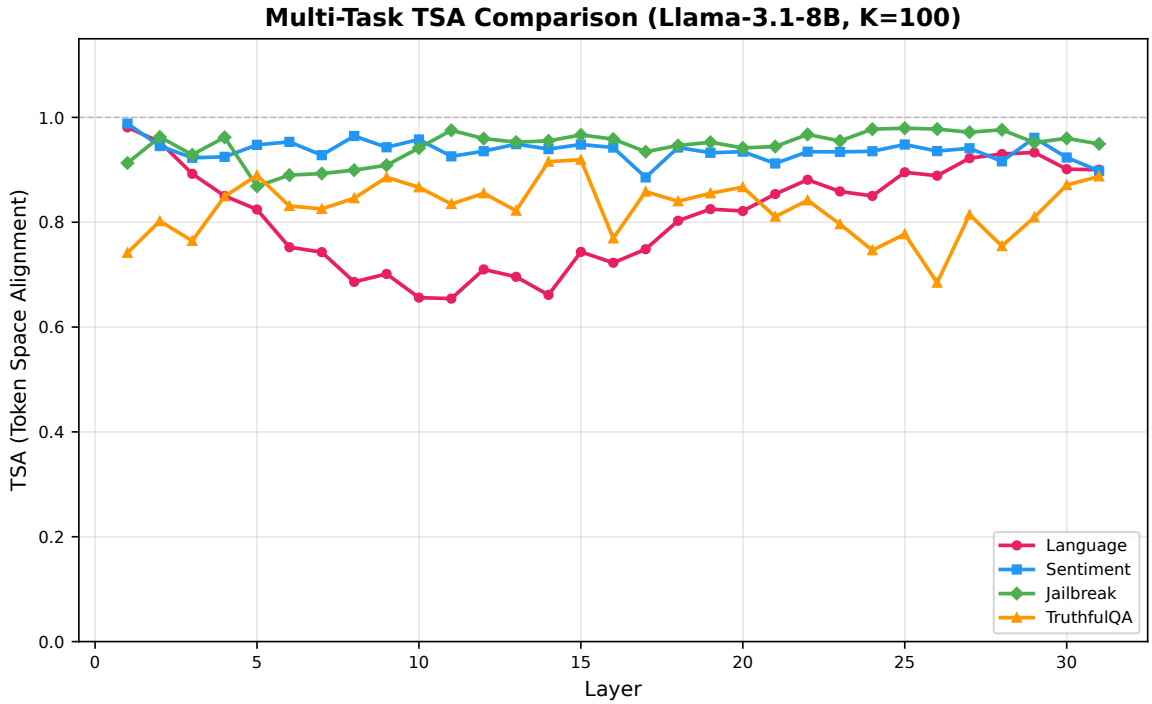

Figure 7: Comparison of TSA curves across four tasks on Llama-3.1-8B at $K = 100$. Language exhibits the strongest U-shape; tasks depending on surface-level properties show the most pronounced token-space departure in middle layers.

D.6 MULTI-TASK TSA COMPARISON

Figure 7 compares the TSA curves of four classification tasks on Llama-3.1-8B at $K = 100$: Language (5-class), Sentiment (2-class), Jailbreak (2-class), and TruthfulQA (2-class). This comparison addresses the question: *how does the nature of the task influence token-space alignment?*

Language ($\mu_{\text{TSA}} = 0.814$, U-score $= +0.145$): The strongest U-shape among all tasks. TSA drops to 0.654 in middle layers (L8–14), losing 34.6% of token-space information. Language classification depends heavily on surface-level properties (script, tokenization patterns, morphemes), and when this surface-level information is abstracted away in middle layers, the token-basis capture rate drops sharply.

Sentiment ($\mu_{\text{TSA}} = 0.937$, U-score $= +0.005$): Nearly flat. Sentiment analysis depends consistently on sentiment vocabulary tokens (e.g., "happy", "terrible") across all layers, so token-space alignment is maintained throughout. The near-absence of a U-shape indicates that sentiment information remains in the token space even during the abstract processing stage.

Jailbreak ($\mu_{\text{TSA}} = 0.946$, U-score $= -0.013$): The highest mean TSA. Jailbreak detection depends strongly on specific pattern tokens (e.g., "ignore", "pretend", "DAN"), and the embedding directions of these tokens stably capture classification information across all layers. Interestingly, TSA is slightly lower in early layers (L1–5, ≈ 0.87), suggesting that jailbreak patterns become clearer after contextual integration.

TruthfulQA ($\mu_{\text{TSA}} = 0.827$, U-score $= -0.046$): The second-lowest mean TSA after Language. However, the pattern is not U-shaped but rather *inverted U-shaped*: TSA peaks in middle layers (L15, 0.919) and reaches its minimum in late layers (L26, 0.685). This suggests that truthfulness classification aligns best with the token space during the semantic integration process in middle layers, but a more abstract judgment process intervenes in the final output layers, causing departure from the token space.

Relationship between task properties and token-space alignment. These results reveal a consistent principle: tasks depending on *surface-level properties* (language, script) exhibit the strongest U-shape, with large departures from token space in middle layers; tasks depending on *lexical properties* (sentiment words, pattern tokens) maintain stable alignment across all layers; and tasks requiring *abstract reasoning* (truthfulness) show maximal alignment in middle layers followed by late-layer departure.

This provides additional evidence for the main-text claim that language information is fundamentally a surface-level, token-space signal: the observation that Language classification shows the strongest U-shape confirms that language cues are token-level surface signals that temporarily become "invisible" during deep abstract processing and are restored at the output stage.

D.7 UNIFIED INTERPRETATION: THREE-PHASE BASIS TRANSITION IN TRANSFORMERS

Synthesizing the five analyses above, the layer-wise processing of transformers can be understood through the lens of *basis transition*:

Table 12: Summary of the three-phase basis transition, with representative metrics from Llama-3.1-8B (Language task, K=100).

Metric	**I: Token-Aligned**	**II: Abstract**	**III: Token-Return**
Representative layers	L1–7	L8–25	L26–31
TSA (K=100)	> 0.85	0.56–0.73	> 0.88
Minimum K (80%)	10–50	500–2,000	50–100
Token diversity	0.60–0.67	0.50–0.55	0.57–0.65
Cross-layer token sim.	High within-block	Low inter-block	High within-block
Interpretation	Vocabulary basis	Abstract basis	Vocabulary basis

This three-phase pattern is precisely symmetric to the "middle-layer token prediction instability" observed in Logit Lens research. Whereas Logit Lens captures this phenomenon via forward projection (hidden state $\rightarrow$ vocabulary), TSA captures it via reverse constraint (vocabulary $\rightarrow$ classification). The convergence of these two independent perspectives carries an important implication: the middle-layer departure from the token space is not an artifact of a particular measurement method but a fundamental structural property of the transformer architecture.

The model *lifts* input tokens into an abstract semantic space, performs reasoning in this space, and then *projects* the results back into the token space for output. This "lifting–processing–projection" cycle provides a geometric basis for the claim that language is an interface for input and output, while the model's core computation occurs in a higher-dimensional abstract representation that transcends the vocabulary. Language information is aligned with the token space in the interface layers (early and late) but transformed into token-transcendent representations in the layers where the model's deep processing takes place.

E HUMAN INTERPRETABILITY & ADDITIONAL RESULTS

E.1 USER STUDY FOR HUMAN INTERPRETABILITY

We conducted a user study to evaluate whether the top-K tokens extracted by Token-Mixture are interpretable to humans. Participants viewed only token lists (without the original input text) and performed two tasks: (1) classify the sample based on the token list alone, and (2) rate the interpretability of the token list on a 5-point Likert scale (1 = Very Low, 5 = Very High).

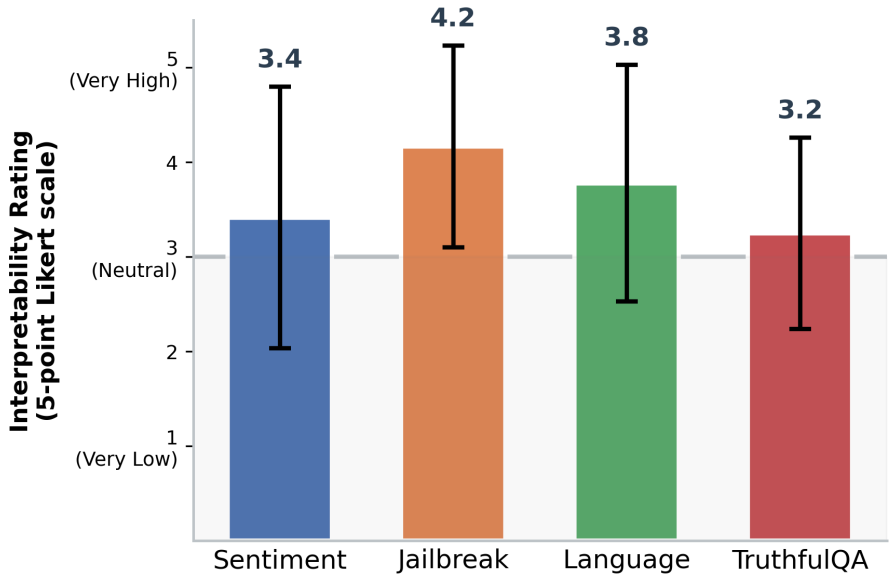

Figure 8: Interpretability ratings on a 5-point Likert scale across four tasks. Error bars indicate ± 1 standard deviation.

Classification accuracy from tokens alone. Participants achieved **100%** accuracy on Jailbreak classification and **92%** on Sentiment classification using only token lists, without access to the original input text. Language identification was considerably harder (25% accuracy), attributable to participants' unfamiliarity with non-English script tokens.

Interpretability ratings. Per-task Likert ratings were: Jailbreak 4.2 ± 1.1, Language 3.8 ± 1.3, Sentiment 3.4 ± 1.4, and TruthfulQA 3.2 (Figure 8). The overall mean rating of 3.83/5 exceeds the neutral midpoint (3.0), confirming that the token lists extracted by Token-Mixture provide a degree of human-interpretable explanation for semantic classification.

Discussion. The high classification accuracy on Jailbreak and Sentiment demonstrates that the selected tokens carry sufficient semantic signal for humans to identify the task-relevant concept. The lower accuracy on Language identification reflects the fact that language-specific tokens (script markers, function words) may not be immediately recognizable to participants unfamiliar with those scripts, rather than a failure of the token selection itself. TruthfulQA received the lowest interpretability rating, consistent with the main-text finding that truthfulness requires broader, more abstract lexical support ($K_{\text{sat}} = 11{,}000$) that is inherently harder to summarize in a compact token list.

E.2 FAILURE CASES

To provide an honest assessment of Token-Mixture's limitations, we identify cases where the selected tokens are difficult to interpret or where the decomposition does not yield clear semantic insight.

Failure type 1: Opaque subword fragments. Several anchor tokens are subword fragments rather than complete words, making their semantic contribution unclear to human readers. For example, in the Sentiment negative direction, tokens such as , `leine`, and `mis` are selected, while the positive direction includes `cro`, `azine`, and `grandi`. These are likely

subword pieces of longer tokens (e.g., "magazine," "grandiose") that carry statistical associations with sentiment patterns in training data, but their isolated appearance in a token list provides limited interpretive value.

Failure type 2: Template and markup artifacts. Tokens such as `.XtraReports`, `gridBagConstraints`, `?>&`, and `.deleteById` appear prominently in the Alignment and Sentiment directions. Similarly, the TruthfulQA directions contain programming artifacts like `.Server`, `mapDispatchToProps`, `Constants`, and `<style`. While these tokens are informative for classification—they may reflect systematic differences in the training data distributions associated with each class—they do not correspond to semantic concepts in the traditional sense. Their selection reveals statistical regularities in the data rather than interpretable semantic structure.

Failure type 3: Cross-lingual ambiguity in language classification. For the Language task, many selected tokens are shared across multiple language directions. For instance, `collaborations` appears in the top-10 of Spanish, Chinese, and German directions simultaneously, and `mud` appears in both Chinese and German. The token `customers` is shared between Spanish and German, while `admitted` appears in both French and German. This cross-class overlap indicates that at Layer 31, the token-level decomposition does not isolate cleanly language-specific signals, as classification relies on distributed patterns across shared tokens rather than exclusive markers.

Failure type 4: Low-data instability (TruthfulQA). TruthfulQA has the smallest training set (1,361 samples), and the per-token weights are an order of magnitude smaller than other tasks (0.07–0.13 vs. 0.4–0.5 for Sentiment and Alignment), indicating that no single token carries strong discriminative power. Some tokens in the TruthfulQA directions (e.g., `Za`, `(visible`,) appear unrelated to truthfulness. This is consistent with the higher K_{sat} for TruthfulQA ($\approx 11{,}000$): the concept is distributed across a wider vocabulary region, making any compact token list an incomplete representation.

Summary. These failure cases highlight that Token-Mixture's interpretability is strongest for tasks with concentrated, lexically grounded signals (Sentiment, Alignment) and weakest for tasks requiring broad, abstract reasoning (TruthfulQA) or where the selected layer yields distributed representations (Language at Layer 31). The presence of subword fragments and programming artifacts suggests that future work could benefit from post-processing steps such as subword merging or token-group clustering to improve human readability.

E.3 REPRESENTATIVE TOKEN LISTS

Table 13 presents the top-10 anchor tokens for each class direction across the four tasks, extracted from Llama-3.1-8B-Instruct at $K = 100$. Tokens are ranked by their contribution weight $\alpha_{i,c}$ in descending order. The layer with the highest validation accuracy is selected independently for each task: Alignment (Layer 25), Sentiment (Layer 10), TruthfulQA (Layer 14), and Language (Layer 31).

Observations. For Alignment, the jailbreak direction contains tokens related to instructions and boundaries (`recognize`, `Instead`, `villain`, `unpleasant`) alongside code artifacts (`\ORM`, `tqdm`), while the normal direction is dominated by neutral vocabulary (`successive`, `Care`, `evident`) and programming terms (`gridBagConstraints`, `.XtraReports`).

For Sentiment, the negative direction includes both indirectly negative tokens (`committee`, `metallic`, `hinges`) and subword fragments (, `mis`, `leine`). Notably, no directly evaluative words like "bad" or "terrible" appear in the top-10, suggesting that the model's sentiment representation is more distributed than lexically grounded. The positive direction similarly mixes semantically suggestive tokens (`Friends`, `grandi`) with opaque fragments (`cro`, `azine`, `LL`).

For Language, the selected tokens at Layer 31 do not function as orthographic markers of each language. Instead, the classification relies on distributed weight patterns: tokens like `collaborations`, `mud`, and `customers` appear in multiple language directions with different weight magnitudes. The only clearly language-specific tokens are `muy` (Spanish), `ères` (French), `Bett`/`usch` (German), and Cyrillic fragments. This distributed pattern is consistent with the observation that K=100 achieves 97% accuracy at Layer 1 but requires more nuanced token combinations at deeper layers.

For TruthfulQA, the false direction contains non-English tokens (,) and template fragments (`/>`, `.null`, `<style`), while the truthful direction contains programming tokens (`.Server`, `mapDispatchToProps`, `Constants`, `BoundingBox`). The per-token weights are notably small (0.07–0.13 vs. 0.4–0.5 for other tasks), confirming that truthfulness is a diffuse concept requiring many tokens to represent. As discussed in §E.2, some of these tokens have limited standalone interpretive value.

Table 13: Representative top-10 anchor tokens per class direction (Llama-3.1-8B-Instruct, K=100). Tokens are ranked by contribution weight $\alpha_{i,c}$. Subword fragments are shown as they appear in the tokenizer. Selected layers: Alignment L25, Sentiment L10, TruthfulQA L14, Language L31.

Task	Class	Top-10 Anchor Tokens (ranked by weight)
Alignment	Jailbreak	recognize, \ORM, tqdm, --from, Non, unpleasant, Malaysia, Instead, villain, numbered
	Normal	quad, 871, successive, gridBagConstraints, mes, .XtraReports, Care, evident, mediate, midfielder
TruthfulQA	False	, U, />, (visible, , Strategy, Za, .null, <style, Cost
	Truthful	This, .Server, wur, mapDispatchToProps, Studios, Constants, ashi, BoundingBox, renovations, _ELEMENTS
Sentiment	Negative	, Instead, modifiers, hinges, mis, committee, metallic, Mayweather, leine, .deleteById
	Positive	Neutral, ?>&, cro, LL, azine, Man, , FTP, grandi, Friends
Language	English	prom, Guatemala, propaganda, [, sliding, well, Richard, occurrence, actively,
	Spanish	-at, restoration, , ascending, circular, collaborations, customers, select, -packed, muy
	Chinese	coaches, muy, Canterbury, mart, mud, irc, collaborations, zo, testified, /reg
	French	irc, , envy, avan, ères, redistribute, logue, Pakistan, admitted, contributed
	German	\Contracts, Bett, 216, customers, mud, collaborations, , admitted, usch, phn